

# Chapter 7 Filter Design Techniques

- Filter – An important class of LTI systems
- We discuss frequency-selective filters mostly: LP, HP, ...
- Three stages in filter design:
  - Specification: application dependent
  - “Design”: approximate the given spec. using a causal discrete-time system
  - Realization: architectures and circuits (IC) implementation
- IIR filter design techniques
- FIR filter design techniques

Frequency domain specifications

Magnitude:  $|H(e^{j\omega})|$

Phase:  $\angle H(e^{j\omega})$

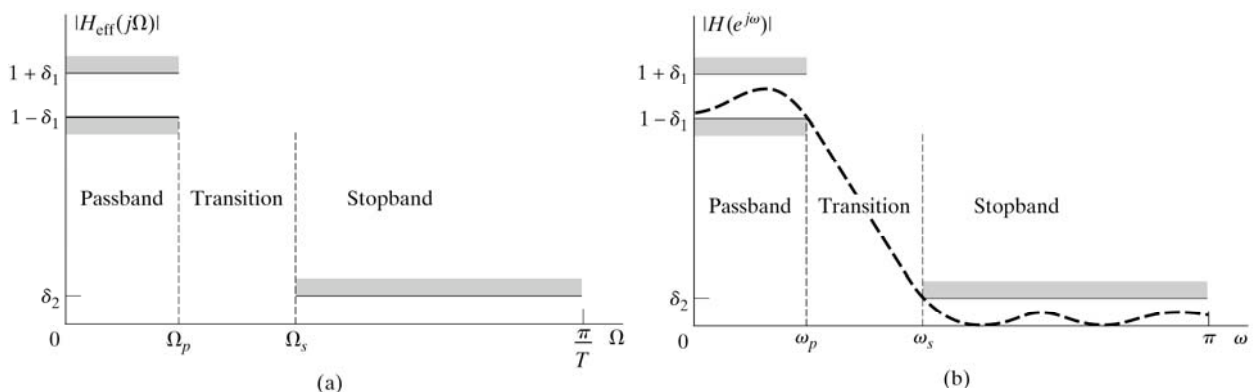
Ex., Low-pass filter: Passband , Transition, Stopband (Fig. 7.1, p.524)

Frequencies: Passband cutoff  $\omega_p$

Stopband cutoff  $\omega_s$

Transition bandwidth  $\omega_s - \omega_p$

Error tolerance  $\delta_1, \delta_2$



**Figure 7.1** Lowpass filter tolerance scheme

## ✧ Analog Filters (Appendix B, p.1080)

### • Butterworth Lowpass Filters

- Monotonic magnitude response in the passband and stopband
- Maximally flat passband

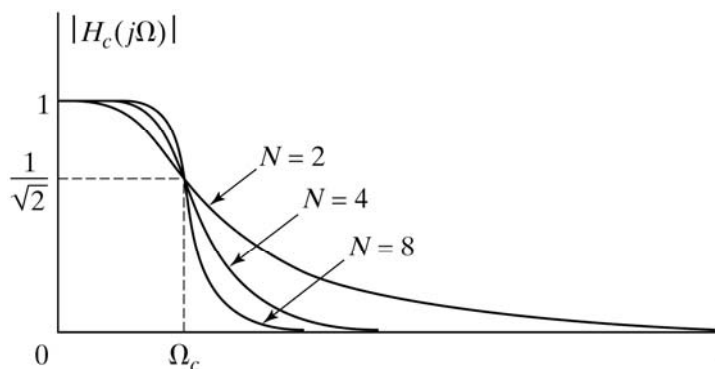
$$|H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{j\Omega}{j\Omega_c}\right)^{2N}}$$

$N$ : filter order

$\Omega_c$ : 3-dB cutoff frequency (magnitude = 0.707)

### ■ Properties

- (a)  $|H_c(j\Omega)|_{\Omega=0} = 1$  (Fig.B.1)
- (b)  $|H_c(j\Omega)|_{\Omega=\Omega_c}^2 = 1/2$  or  $|H_c(j\Omega)|_{\Omega=\Omega_c} = 0.707$  (Fig.B.2)
- (c) The first  $(2N-1)$  derivatives of  $|H_c(j\Omega)|^2$  are zero at  $\Omega = 0$   
 ➔ maximum flat
- (d)  $|H_c(j\Omega)|^2$  is monotonically decreasing (of  $\Omega$ )
- (e)  $N \rightarrow \infty \rightarrow |H_c(j\Omega)| \rightarrow$  ideal lowpass



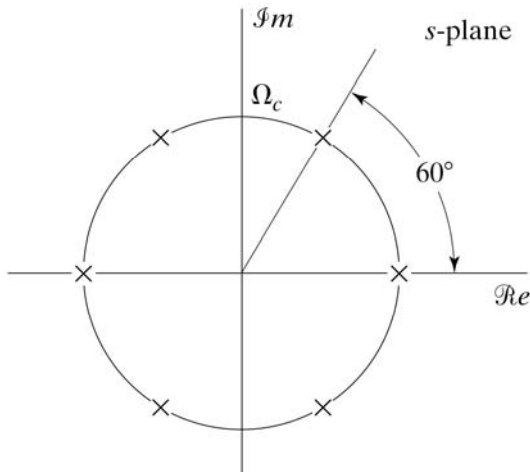
**Figure B.2** Dependence of Butterworth magnitude characteristics on the order  $N$ .

### ■ Poles

$$H_c(s)H_c(-s) = \frac{1}{1 + \left(\frac{s}{j\Omega_c}\right)^{2N}}$$

$$\text{Roots: } s_k = (-1)^{\frac{1}{2N}} (j\Omega_c) = \Omega_c e^{j\frac{\pi}{2N}(2k+N-1)}, \quad k = 0, 1, \dots, 2N-1$$

- (a)  $2N$  poles in pairs:  $s_k, -s_k$  symmetric w.r.t. the imaginary axis;  
never on the imaginary axis. If  $N$  odd, poles on the real axis.
- (b) Equally spaced on a circle of radius  $\Omega_c$  (Fig.B.3)
- (c)  $H_c(s)$  causal, stable  $\leftarrow$  all poles on the left half plane



**Figure B.3** s-plane pole locations for a third-order Butterworth filter.

■ Usage (There are only two parameters  $N, \Omega_c$ )

Given specifications  $\varepsilon, \Omega_p, \delta_2, \Omega_s \rightarrow N, \Omega_c$

$$|H(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}} = \frac{1}{1 + \varepsilon^2 \left(\frac{\Omega}{\Omega_p}\right)^{2N}}$$

$$\text{Thus, } |H(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2} \quad \text{at} \quad \Omega = \Omega_p \Rightarrow \Omega_c = \frac{\Omega_p}{\varepsilon^{\frac{1}{N}}}$$

$$\text{At } \Omega = \Omega_s, \quad |H(j\Omega)|_{\Omega_s}^2 = \delta_2^2 = \frac{1}{1 + \varepsilon^2 \left(\frac{\Omega_s}{\Omega_p}\right)^{2N}}$$

$$N = \frac{\log\left[\left(\frac{1}{\delta_2}\right)^2 - 1\right]}{2\log\left(\frac{\Omega_s}{\Omega_c}\right)}$$

- **Chebyshev Filters**

- **Type I:** Equiripple in the passband; monotonic in the stopband

- Type II:** Equiripple in the stopband; monotonic in the passband

- Same  $N$  as the Butterworth filter, it would have a sharper transition band. (A smaller  $N$  would satisfy the spec.)

- **Type I:**

$$|H_c(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 V_N^2(\Omega/\Omega_c)}$$

where  $V_N(x)$  is the  $N$ th-order Chebyshev polynomial

$$V_N(x) = \cos(N \cos^{-1}(x)), \quad 0 < V_N(x) < 1 \text{ for } 0 < x < 1$$

$$V_{N+1}(x) = 2xV_N(x) - V_{N-1}(x)$$

$$V_N(x)|_{x=1} = 1 \text{ for all } N$$

**<The first several Chebyshev polynomials>**

| $N$      | $V_N(x)$          |
|----------|-------------------|
| <b>0</b> | 1                 |
| <b>1</b> | $x$               |
| <b>2</b> | $2x^2 - 1$        |
| <b>3</b> | $4x^3 - 3x$       |
| <b>4</b> | $8x^4 - 8x^2 + 1$ |

- Properties (Type I)

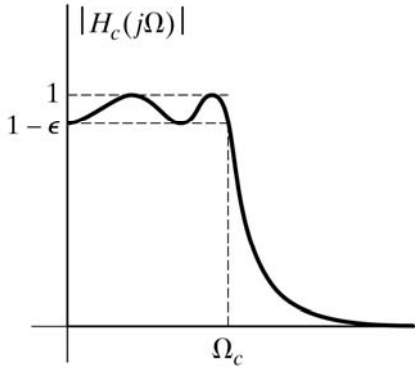
$$(a) |H_c(j\Omega)|_{\Omega=0}^2 = \begin{cases} 1, & \text{if } N \text{ odd} \\ \frac{1}{1 + \varepsilon^2}, & \text{if } N \text{ even} \end{cases}$$

(b) The magnitude squared frequency response oscillates between

1 and  $\frac{1}{1+\varepsilon^2}$  within the passband:

$$|H_c(j\Omega)|_{\Omega=\Omega_c}^2 = \frac{1}{1+\varepsilon^2} \quad \text{at } \Omega = \Omega_c$$

(c)  $|H_c(j\Omega)|^2$  is monotonic outside the passband.



**Figure B.4** Type I Chebyshev lowpass filter approximation.

#### ■ Poles (Type I)

On the ellipse specified by the following: (Fig.B.5)

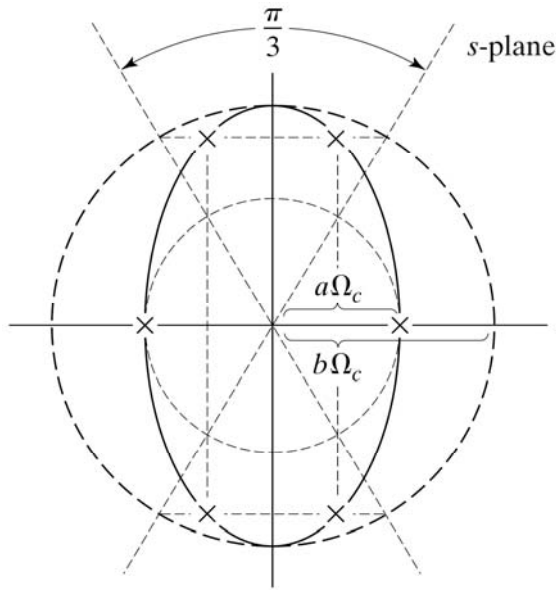
$$\text{Length of minor axis} = 2a\Omega_c, \quad a = \frac{1}{2} \left( \alpha^{\frac{1}{N}} - \alpha^{-\frac{1}{N}} \right)$$

$$\text{Length of major axis} = 2b\Omega_c, \quad b = \frac{1}{2} \left( \alpha^{\frac{1}{N}} + \alpha^{-\frac{1}{N}} \right)$$

$$\text{and } \alpha = \varepsilon^{-1} + \sqrt{1 + \varepsilon^{-2}}$$

(a) Locate equal-spaced points on the major circle and minor circle with angle  $\Phi_k = \frac{\pi}{2} + \frac{(2k+1)\pi}{N}, k = 0, 1, \dots, N-1$

(b) The poles are  $(x_k, y_k)$ :  $x_k = a\Omega_c \cos \phi_k, y_k = b\Omega_c \sin \phi_k$



**Figure B.5** Location of poles for a third-order type I lowpass Chebyshev filter.

■ **Type II:**

$$|H_a(j\Omega)|^2 = \frac{1}{1 + [\varepsilon^2 V_N^2(\Omega_c/\Omega)]^{-1}}$$

has both poles and zeros.

■ **Usage** (There are only two parameters  $N, \Omega_c$ )

Given specifications  $\varepsilon, \Omega_p, \delta_2, \Omega_s \rightarrow N, \Omega_c$

$$\Omega_c = \Omega_p$$

$N=?$  “Complicated!”

$$N = \frac{\log[(\sqrt{1 - \delta_2^2} + \sqrt{1 - \delta_2^2(1 + \varepsilon^2)})/\varepsilon\delta_2]}{\log[(\Omega_s/\Omega_p) + \sqrt{(\Omega_s/\Omega_p)^2 - 1}]}$$

$$= \frac{\cosh^{-1}(\delta/\varepsilon)}{\cosh^{-1}(\Omega_s/\Omega_p)} \quad \left( \delta_2 = \frac{1}{\sqrt{1 + \delta^2}} \right)$$

- **Elliptic Filters**

- Equiripple at both the passband and the stopband (Fig.B.6)
- Optimum: smallest  $(\Omega_s - \Omega_p)$  at the same  $N$

$$|H_a(j\Omega)|^2 = \frac{1}{1 + \varepsilon^2 U_N^2(\Omega/\Omega_p)}$$

where  $U_N(x)$ : Jacobian elliptic function (Very complicated! Skip!)

- Usage (There are only two parameters  $N, \Omega_c$ )

Given specifications  $\varepsilon, \Omega_p, \delta_2, \Omega_s \rightarrow N, \Omega_c$

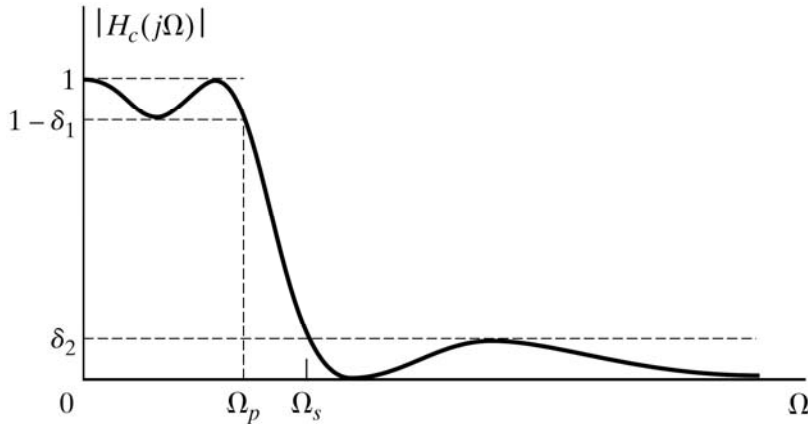
$$N = \frac{K(\Omega_p/\Omega_s)K(\sqrt{1 - (\varepsilon^2/\delta^2)})}{K(\varepsilon/\delta)K(\sqrt{1 - (\Omega_p/\Omega_s)^2})} \quad \left( \delta_2 = \frac{1}{\sqrt{1 + \delta^2}} \right)$$

where  $K(x)$  is the complete elliptic integral of the first kind

$$K(x) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - x^2 \sin^2 \theta}}$$

Using computer package to design.

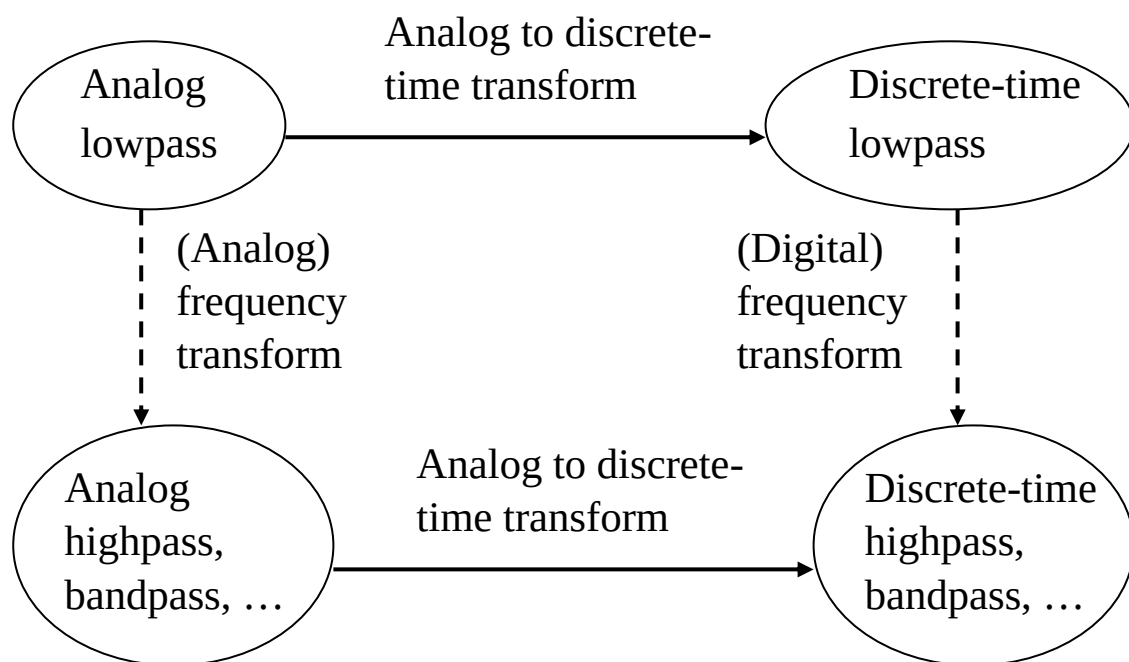
*Remark:* The drawback of the elliptic filters: They have more non-linear phase response in the passband than a comparable Butterworth filter or a Chebyshev filter, particularly, near the passband edge.



**Figure B.6** Equiripple approximation in both passband and stopband.

## ✧ Design Digital IIR Filters from Analog Filters

- Why based on analog filters?
  - Analog filter design methods have been well developed.
  - Analog filters often have simple *closed-form* design formulas.
    - ✦ Direct digital filter design methods often don't have *closed-form* formulas.
- There are two types of transformations
  - Transformation from analog to discrete-time
  - Transformation from one type filter to another type (so called *frequency transformation*)



- Methods in analog to discrete-time transformation
  - Impulse invariance
  - Bilinear transformation
  - Matched-z transformation



- Desired properties of the transformations
  - Imaginary axis of the s-plane  $\rightarrow$  The unit circle of the z-plane
  - Stable analog system  $\rightarrow$  Stable discrete-time system  
(Poles in the left s-plane  $\rightarrow$  Poles inside the unit circle)
- Steps in the design
  - (1) Digital specifications  $\rightarrow$  Analog specifications
  - (2) Design the desired analog filter
  - (3) Analog filter  $\rightarrow$  Discrete-time filter

- **Impulse Invariance**

-- Sampling the impulse of a continuous-time system

$$\begin{aligned} h[n] &= T_d h_c(nT_d) \\ &= T_d h_c(t) \big|_{t=nT_d} \end{aligned}$$

$T_d$ : Sampling period

$\leftarrow$  *Important:* to avoid aliasing

$\leftarrow$  Does not show up in the final discrete formula if we start from the digital specifications, ...

- Frequency response

Sampling in time  $\rightarrow$  Sifted duplication in frequency

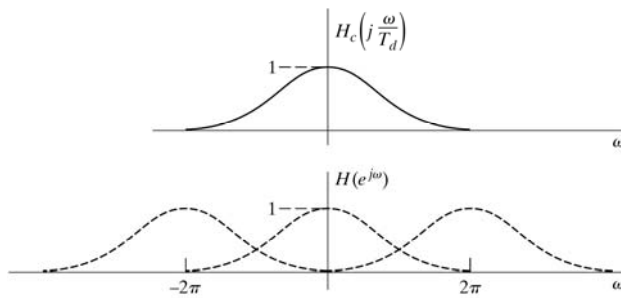
$$H(e^{j\omega}) = \sum_{k=-\infty}^{\infty} H_c(j\frac{\omega}{T_d} + j\frac{2\pi}{T_d}k)$$

If  $H_c(j\Omega)$  is band-limited and  $f_d = 1/T_d$  is higher than the

Nyquist sampling frequency (no aliasing)

$$H(e^{j\omega}) = H_c(j\frac{\omega}{T_d}) \quad | \omega | \leq \pi$$

*Remark:* This is not possible because the IIR analog filter is typically not bandlimited.



**Figure 7.3** Illustration of aliasing in the impulse invariance design technique.

Here are two approaches:

*Approach 1:* Sampling  $h[n]$

*Approach 2:* Map  $H_c(s)$  to  $H(z)$  because we need  $H(z)$  to implement a digital filter anyway.

$$H_c(s) = \sum_{k=1}^N \frac{A_k}{s - s_k}$$

$$h_c(t) = \begin{cases} \sum_{k=1}^N A_k e^{s_k t}, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

$$h[n] = T_d h_c(nT_d)$$

$$= T_d \sum_{K=1}^N A_k e^{s_k n T_d} u[n]$$

$$= \sum_{K=1}^N (T_d A_k) (e^{s_k T_d})^n u[n]$$

$$H(z) = \sum_{K=1}^N \frac{T_d A_k}{1 - e^{s_k T_d} z^{-1}}$$

Essentially, **factorize and map:**

**Analog pole**

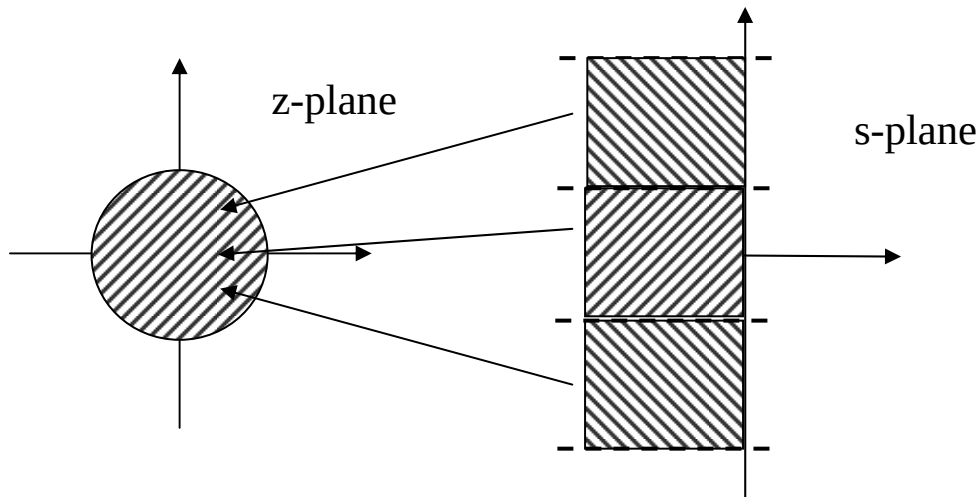
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**Discrete-time pole**

Remarks: (1) Stability is preserved:

LHS poles  $\rightarrow$  poles inside the unit circle

(2) No simple correspondence for zeros



*Design Example:* (O&S, p.529) Low-pass filter

Using Butterworth continuous-time filter

Given specifications in the digital domain

“-1 dB in passband” and “-15 dB in stopband”

$$0.89125 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.17783, \quad 0.3\pi \leq \omega \leq \pi$$

*Step 1:* Convert the above specifications to the analog domain

(Assume “negligible aliasing”)

$$H(e^{j\omega}) = H_c(j\frac{\omega}{T_d}) \quad |\omega| \leq \pi$$

$$0.89125 \leq |H(j\Omega)| \leq 1, \quad 0 \leq \Omega \leq 0.2\pi/T_d$$

$$|H(j\Omega)| \leq 0.17783, \quad 0.3\pi/T_d \leq \Omega \leq \pi/T_d$$

*Step 2:* Design a Butterworth filter that satisfies the above speci-

fications. That is, select proper  $N, \Omega_c$ .

$$\begin{cases} |H_c(j\frac{0.2\pi}{T_d})| \geq 0.89125 \\ |H_c(j\frac{0.3\pi}{T_d})| \leq 0.17783 \end{cases}$$

$$|H_c(j\Omega)|^2 = \frac{1}{1 + (\Omega/\Omega_c)^{2N}}$$

$$\text{Thus, } \begin{cases} 1 + \left(\frac{0.2\pi}{T_d\Omega_c}\right)^{2N} = \left(\frac{1}{0.89125}\right)^2 \\ 1 + \left(\frac{0.3\pi}{T_d\Omega_c}\right)^{2N} = \left(\frac{1}{0.17783}\right)^2 \end{cases}$$

$$\rightarrow N = 5.8858, \quad T_d\Omega_c = 0.70474$$

$\rightarrow$  (Taking integer)  $N = 6, \quad T_d\Omega_c = 0.7032$  (Meet passband spec. exactly; overdesign at stopband)

$$\text{<Case 1: Assume } T_d = 1\text{> } s_k = \Omega_c e^{j\frac{\pi}{2N}(2k+N-1)}$$

$$\text{<Case 2: Assume } T_d \neq 1\text{> } s_k = \left(\frac{0.7032}{T_d}\right) e^{j\frac{\pi}{2N}(2k+N-1)}$$

$$H_c(s) = \frac{0.12093}{(s^2 + 0.365s + 0.495)(s^2 + 0.995s + 0.495)(s^2 + 1.359s + 0.495)}$$

Step 3: Convert analog filter to discrete-time

**Analog pole**  $s_k$

$\Downarrow$

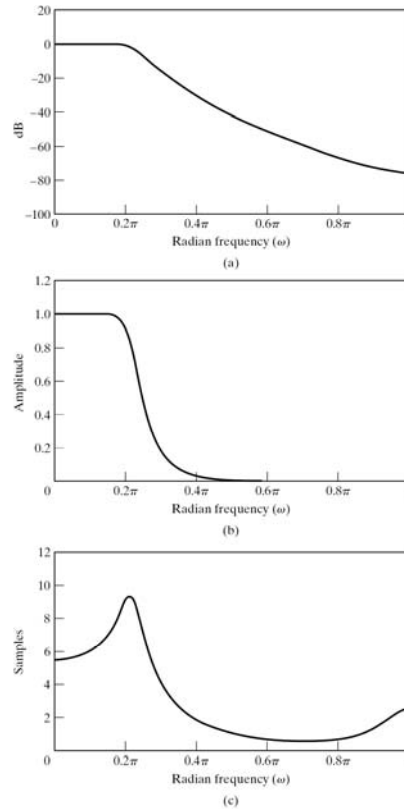
**Discrete-time pole**  $e^{s_k}$

$$\langle \text{Case 1: Assume } T_d = 1 \rangle \quad z_k = \exp \left[ 0.7032 e^{j \frac{\pi}{2N} (2k+N-1)} \right]$$

$$\langle \text{Case 2: Assume } T_d \neq 1 \rangle \quad z_k = \exp \left[ T_d \left( \frac{0.7032}{T_d} \right) e^{j \frac{\pi}{2N} (2k+N-1)} \right]$$

*They are identical! (In general, this is true.)*

$$H(z) = \frac{0.287 - 0.447z^{-1}}{1 - 1.297z^{-1} + 0.695z^{-2}} + \frac{-2.143 + 1.145z^{-1}}{1 - 1.069z^{-1} + 0.370z^{-2}} + \frac{1.856 - 0.630z^{-1}}{1 - 0.997z^{-1} + 0.257z^{-2}}$$



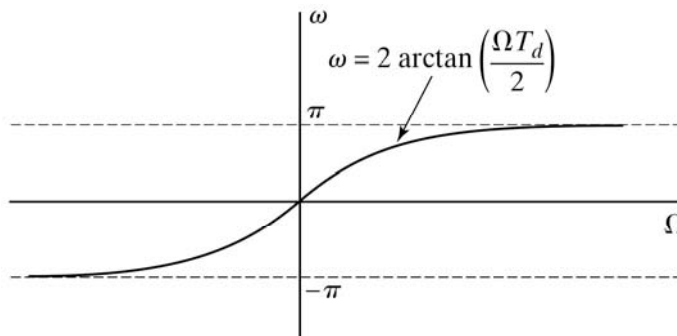
**Figure 7.5** Frequency response of sixth-order Butterworth filter transformed by impulse invariance. (a) Log magnitude in dB. (b) Magnitude. (c) Group delay.

**Remark:** (1) Impulse invariance method has a precise control on the shape of the time signal.

(2) Except for aliasing, the shape of the frequency response is preserved.

- **Bilinear Transform**

- Avoid aliasing but distort the frequency response – uneven stretch of the frequency axis.



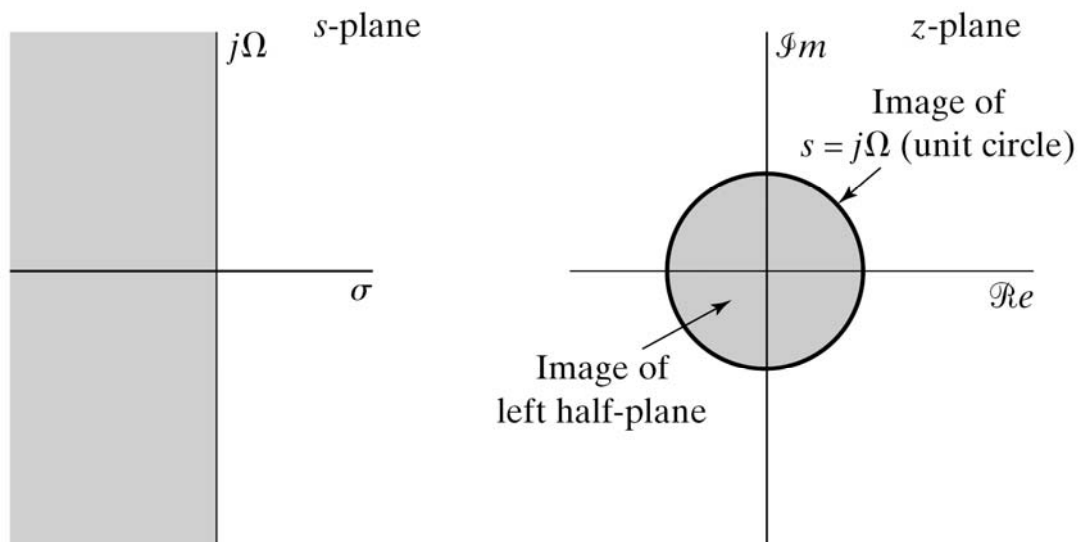
**Figure 7.7** Mapping of the continuous-time frequency axis onto the discrete-time frequency axis by bilinear transformation.

- $$s = \frac{2}{T_d} \left( \frac{1 - z^{-1}}{1 + z^{-1}} \right) \quad \text{or} \quad z = \frac{1 + sT_d/2}{1 - sT_d/2}$$

$$H_c(s) \rightarrow H(z) = H_c \left( \frac{2}{T_d} \left( \frac{1 - z^{-1}}{1 + z^{-1}} \right) \right)$$

*Note:*  $j\Omega$  axis on the s-plane  $\rightarrow$  unit circle on the z-plane (P. 535)

LHS of the s-plane  $\rightarrow$  Interior of the unit circle on the z-plane

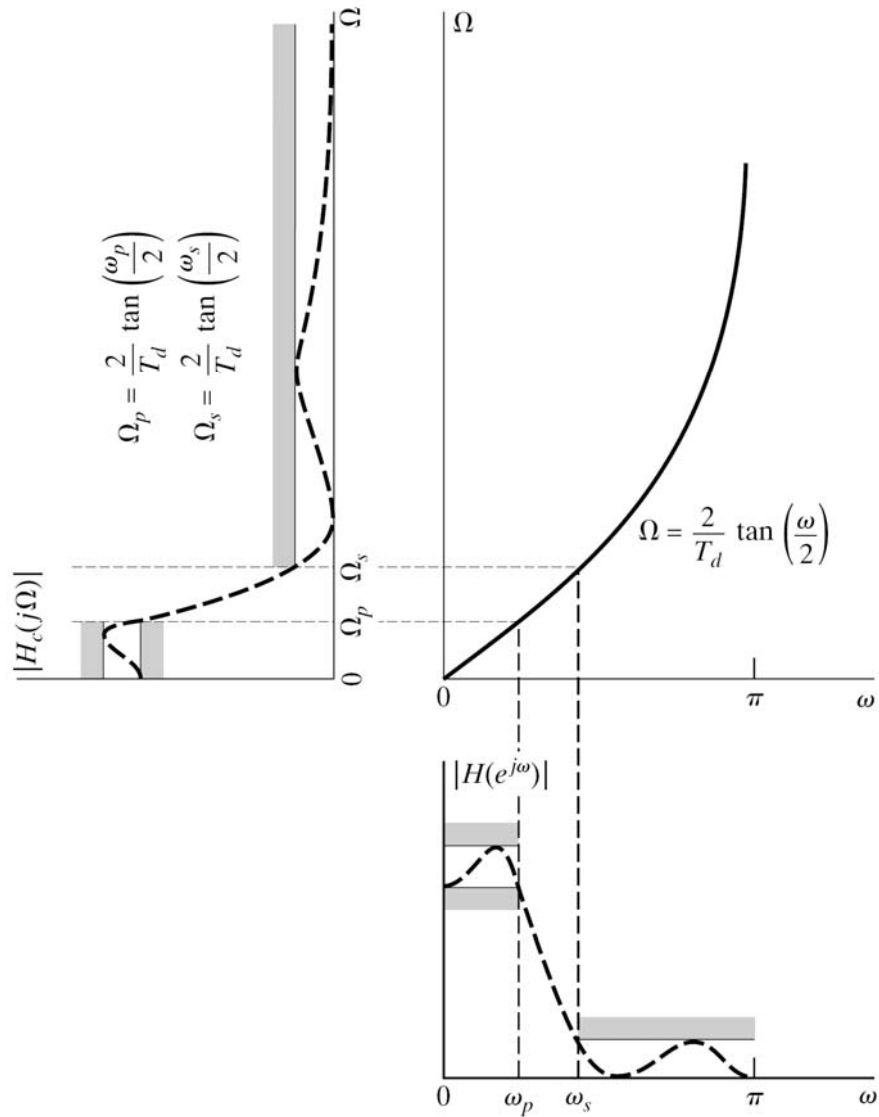


**Figure 7.6** Mapping of the s-plane onto the z-plane using the bilinear transformation.

- How the  $j\Omega$  axis is mapped to the unit circle? (Fig. 7.8, P.536)

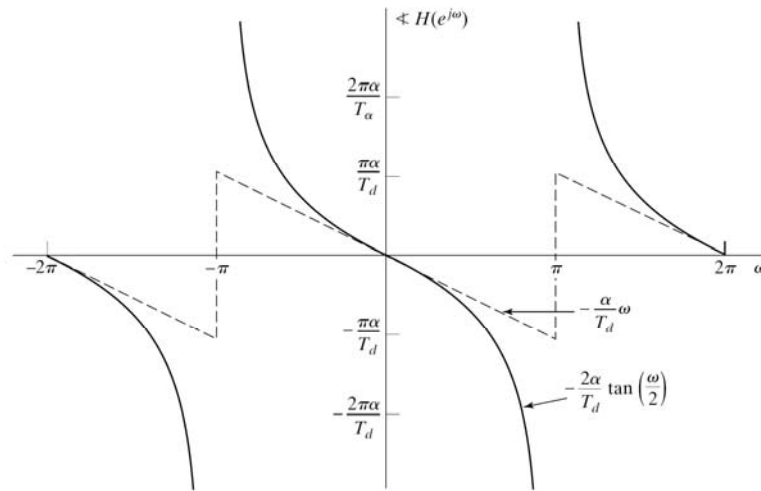
$$s = \frac{2}{T_d} \left( \frac{1-z^{-1}}{1+z^{-1}} \right) \bigg|_{z=e^{j\omega}} = \frac{2}{T_d} \left( \frac{1-e^{-j\omega}}{1+e^{-j\omega}} \right) = \sigma + j\Omega = \frac{2}{T_d} \left[ \frac{2e^{-j\omega/2} \left( j \sin \frac{\omega}{2} \right)}{2e^{-j\omega/2} \left( \cos \frac{\omega}{2} \right)} \right] = \frac{2j}{T_d} \tan \left( \frac{\omega}{2} \right)$$

$$\Rightarrow \Omega = \frac{2}{T_d} \tan \left( \frac{\omega}{2} \right) \text{ or } \omega = 2 \tan^{-1} \left( \frac{\Omega T_d}{2} \right)$$



**Figure 7.8** Frequency warping inherent in the bilinear transformation of a continuous-time lowpass filter into a discrete-time lowpass filter. To achieve the desired discrete-time cutoff frequencies, the continuous-time cutoff frequencies must be prewarped as indicated.

*Problem in design – nonlinear distortion in magnitude and phase*  
(Fig. 7.9, P.537)



**Figure 7.9** Illustration of the effect of the bilinear transformation on a linear phase characteristic. (Dashed line is linear phase and solid line is phase resulting from bilinear transformation.)

### ■ Steps in the design

- (1) Digital specifications to analog specifications: prewarp
- (2) Design the desired analog filter
- (3) Analog filter to discrete-time filter: bilinear transform

### ■ *Design Example:* (O&S, P.538) Lowpass filter

Using Butterworth continuous-time filter

Given specifications in the digital domain (same as the previous ex.)

$$0.89125 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq 0.2\pi$$

$$|H(e^{j\omega})| \leq 0.17783, \quad 0.3\pi \leq \omega \leq \pi$$

$$\text{Step 1: Prewarp } \Omega = \frac{2}{T_d} \tan\left(\frac{\omega}{2}\right)$$

$$\text{Passband freq. } \Omega_p = \frac{2}{T_d} \tan\left(\frac{0.2\pi}{2}\right)$$

$$\text{Stopband freq. } \Omega_s = \frac{2}{T_d} \tan\left(\frac{0.3\pi}{2}\right)$$

Let  $T_d = 1$  since  $T_d$  will disappear after “analog to discrete”.



Step 2: Design a Butterworth filter -- select proper  $N, \Omega_c$ .

$$\begin{cases} |H_c(j2 \tan(0.1\pi))| \geq 0.89125 \\ |H_c(j2 \tan(0.15\pi))| \leq 0.17783 \end{cases}$$

$$\text{Because } |H_c(j\Omega)|^2 = \frac{1}{1 + \left(\frac{\Omega}{\Omega_c}\right)^{2N}}$$

$$\rightarrow \begin{cases} 1 + \left(\frac{2 \tan(0.1\pi)}{\Omega_c}\right)^{2N} = \left(\frac{1}{0.89125}\right)^2 \\ 1 + \left(\frac{2 \tan(0.15\pi)}{\Omega_c}\right)^{2N} = \left(\frac{1}{0.17783}\right)^2 \end{cases}$$

$$\rightarrow N = 5.30466,$$

$$\rightarrow N = 6, \quad T_d \Omega_c = 0.76622 \text{ (Meet stopband spec. exactly; exceed passband spec.)}$$

$$H_c(s) = \frac{0.20238}{(s^2 + 0.3996s + 0.5871)(s^2 + 1.0836s + 0.5871)(s^2 + 1.4802s + 0.5871)}$$

Step 3: Convert analog filter to discrete-time

$$H_c(s) \rightarrow H(z) = H_c\left(2\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\right)$$

$$\begin{aligned} H(z) &= \frac{0.0007378(1+z^{-1})^6}{(1-1.2686z^{-1}+0.7051z^{-2})(1-1.0106z^{-1}+0.3583z^{-2})} \\ &\quad \times \frac{1}{(1-0.9044z^{-1}+0.2155z^{-2})} \end{aligned}$$

*Remarks:* (1) Bilinear transforms warps frequency values but preserves the magnitude. Therefore, the discrete-time Butterworth filter still has the maximal flat property; Chebyshev and Elliptic

filters have equal ripple property.

(2) Although we may obtain  $H_c(s)$  in closed form, it is often difficult to find the locations of poles and zeros of  $H(z)$  from  $H_c(s)$  directly.

■ *Bilinear Transform Design Example using 4 analog filters: (O&S, P.543-548)*

Passband edge frequency  $\omega_p = 0.5\pi$

Stopband edge frequency  $\omega_s = 0.6\pi$

Maximum passband gain= 0dB

Minimum passband gain= -0.3dB

Minimum stopband gain= -30dB

$$20\log_{10}(1+\delta_{p1}) = 0 \quad \text{or equivalently} \quad \delta_{p1} = 0$$

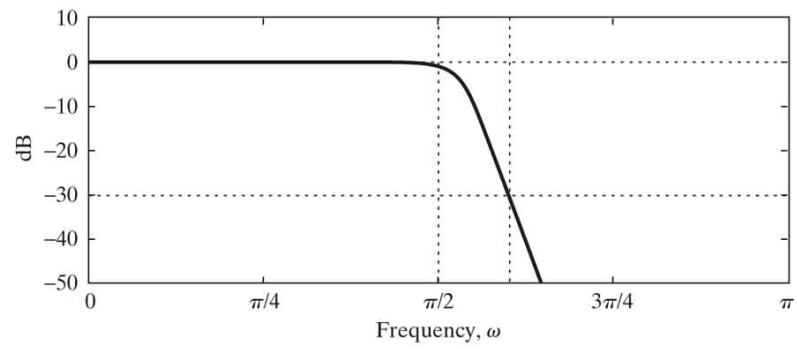
$$20\log_{10}(1-\delta_{p2}) = -0.3 \quad \text{or equivalently} \quad \delta_{p2} = 0.0339$$

$$20\log_{10}(1-\delta_s) = -30 \quad \text{or equivalently} \quad \delta_s = 0.0316$$

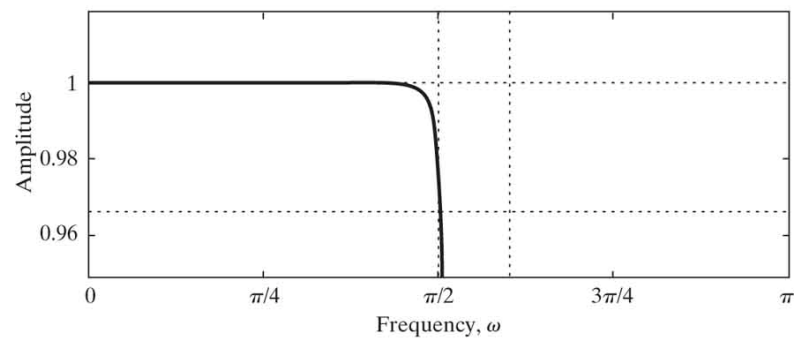
Butterworth: 15<sup>th</sup> order

Chebyshev I and II: 7<sup>th</sup> order

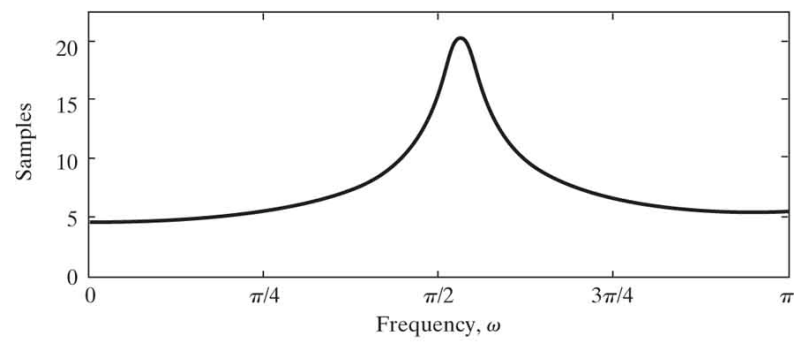
Elliptic: 5<sup>th</sup> order



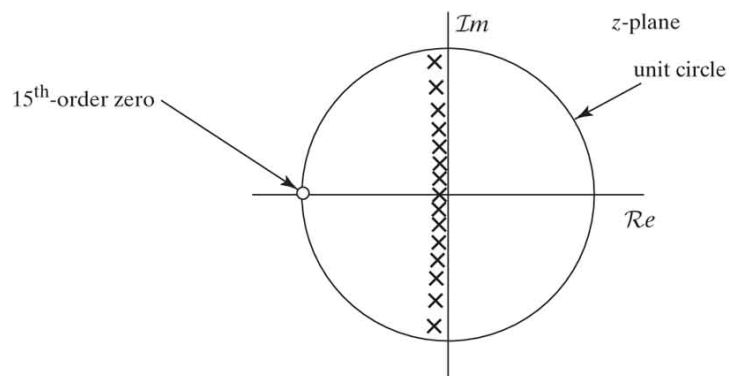
(a)



(b)

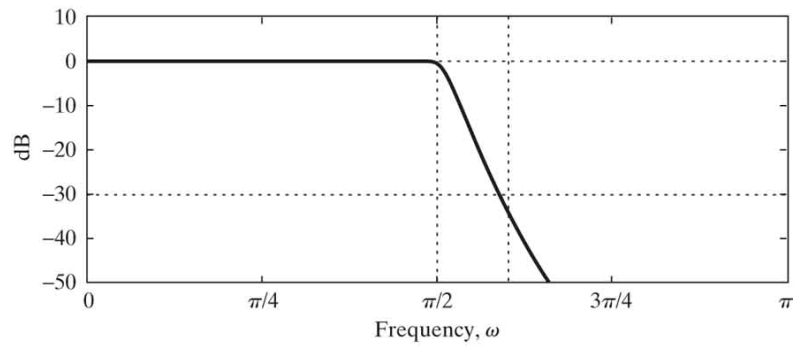


(c)

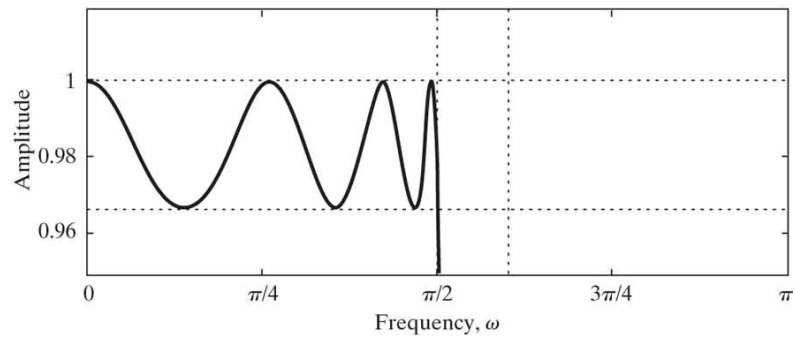


(d)

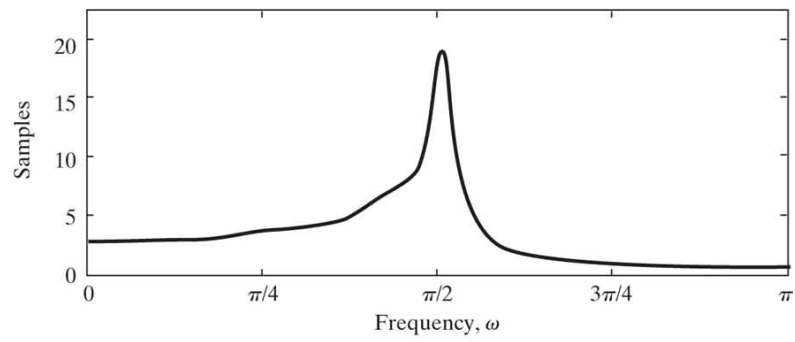
Figure 7.12 Butterworth filter, 15<sup>th</sup>-order.



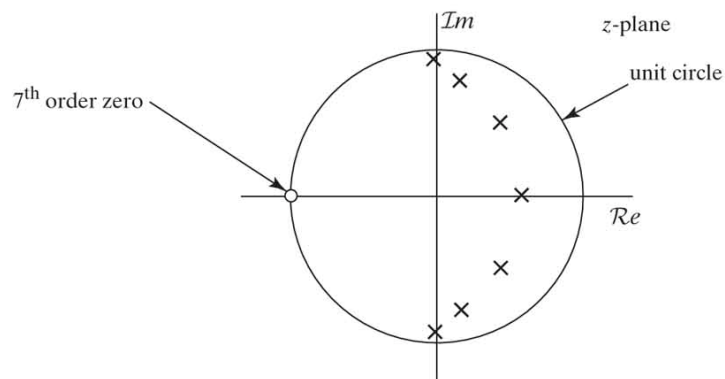
(a)



(b)



(c)



(d)

Figure 7.13 Chebyshev Type I filter, 7<sup>th</sup>-order.

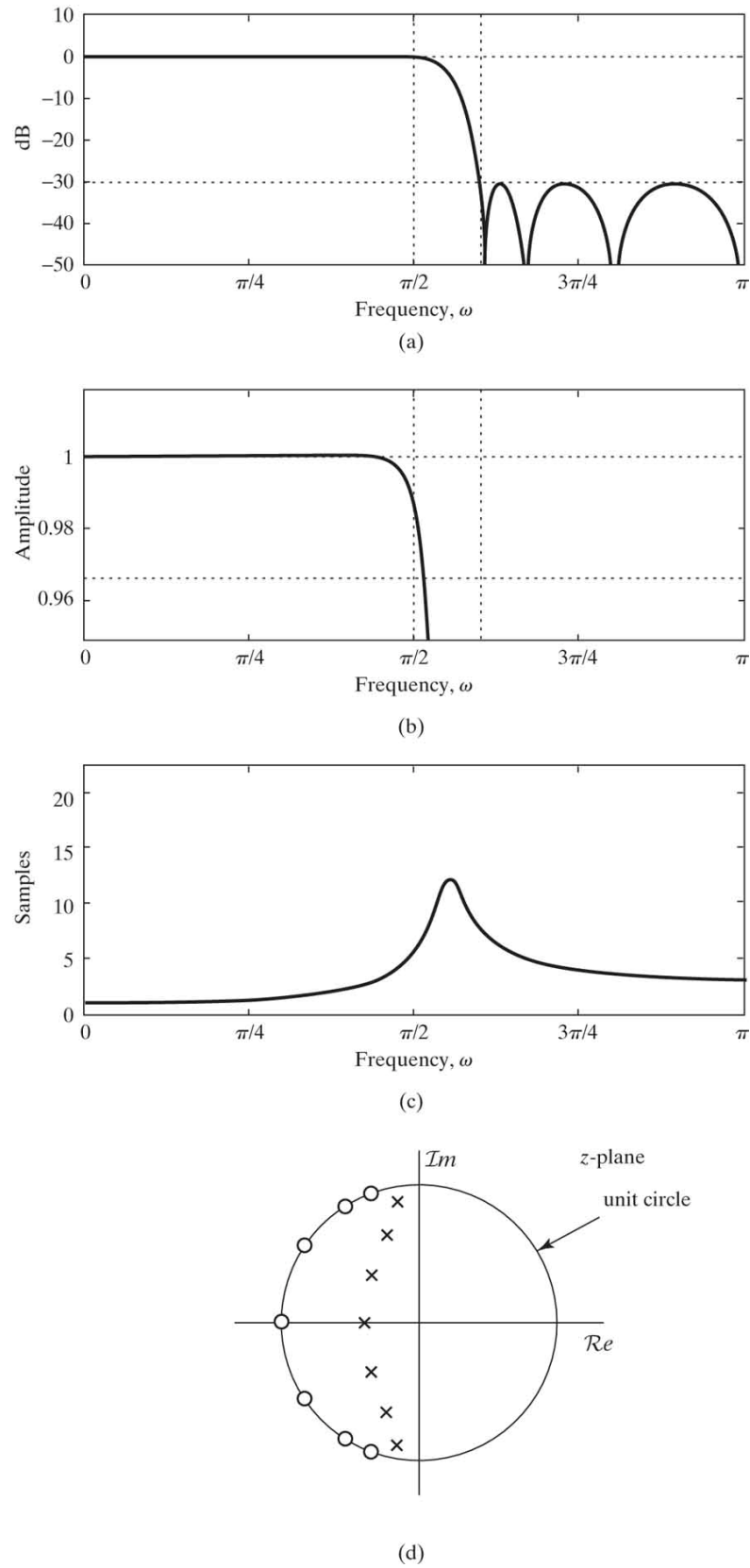


Figure 7.14 Chebyshev Type II filter, 7<sup>th</sup>-order.

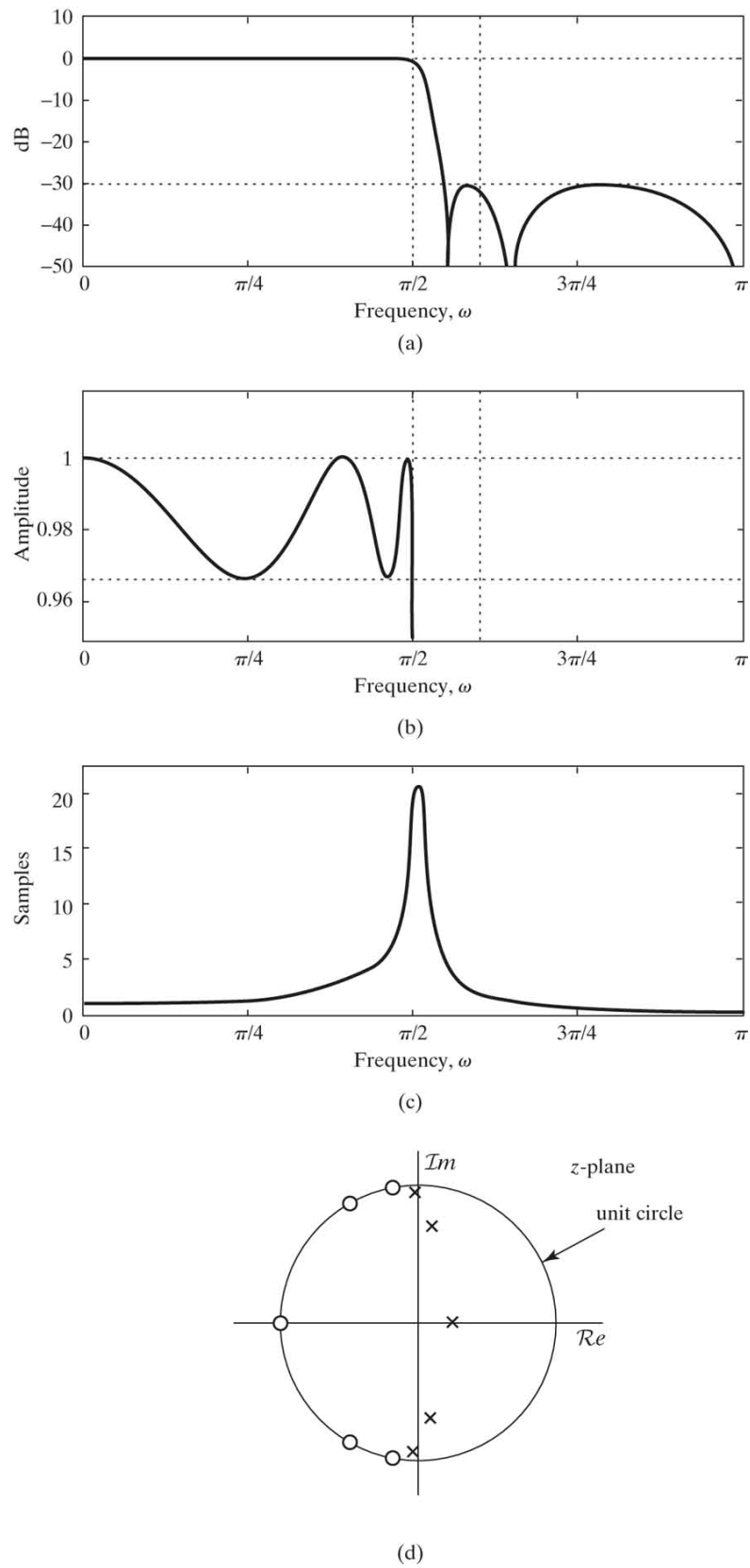


Figure 7.15 Elliptic filter, 5<sup>th</sup>-order, exceeds design specifications.

- **Frequency Transformation**

-- Transform one-type (often lowpass) filter to another type.

Typically, we first design a *frequency-normalized prototype low-pass* filter. Then, use an algebraic transformation to derive the desired lowpass, high pass , ..., filters from the prototype lowpass filter.

<Prototype filter>       $\rightarrow$       <Desired filter>

$Z$                        $\rightarrow$                        $z$

$$Z^{-1} = G(z^{-1})$$

$$H_{lp}(Z) \Big|_{Z^{-1}=G(z^{-1})} \rightarrow H(z)$$

Typically, this transform is made of all-pass like factors

$$G(z^{-1}) = \pm \prod_{K=1}^N \left( \frac{z^{-1} - \alpha_k}{1 - \alpha_k z^{-1}} \right)$$

*Remarks:* The desired properties of  $G(\cdot)$  are (1) transforms the unit circle in  $Z$  to the unit circle in  $z$ , (2) transforms the interior of the unit circle in  $Z$  to the interior of the unit circle in  $z$ , and (3)  $G(\cdot)$  is rational.

*Example:* Lowpass to lowpass (with different passband and stopband frequency, but magnitude is not changed)

$$Z^{-1} = \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$$

Check the relationship between  $\theta$  (the  $Z$  filter) and  $\omega$  (the  $z$  filter).

$\alpha$  is a parameter. Different  $\alpha$  offers different “shapes” of the transformed filters in  $\omega$ .

$$e^{-j\theta} = \frac{e^{-j\omega} - \alpha}{1 - \alpha e^{-j\omega}}$$

$$\omega = \tan^{-1} \left[ \frac{(1 - \alpha^2) \sin \theta}{2\alpha + (1 + \alpha^2) \cos \theta} \right]$$

If  $\theta_p$  is to be mapped to  $\omega_p$ , then

$$\alpha = \frac{\sin[(\theta_p - \omega_p)/2]}{\sin[(\theta_p + \omega_p)/2]}$$

### ■ Various Digital to Digital Transformations

| Filter Type | Transformation                                                                                                                        | Associated Design Formulas                                                                                                                                                                                                                                                                                                      |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Lowpass     | $Z^{-1} = \frac{z^{-1} - \alpha}{1 - \alpha z^{-1}}$                                                                                  | $\alpha = \frac{\sin\left(\frac{\theta_p - \omega_p}{2}\right)}{\sin\left(\frac{\theta_p + \omega_p}{2}\right)}$ $\omega_p = \text{desired cutoff freq.}$                                                                                                                                                                       |
| Highpass    | $Z^{-1} = -\frac{z^{-1} + \alpha}{1 + \alpha z^{-1}}$                                                                                 | $\alpha = -\frac{\cos\left(\frac{\theta_p + \omega_p}{2}\right)}{\cos\left(\frac{\theta_p - \omega_p}{2}\right)}$ $\omega_p = \text{desired cutoff freq.}$                                                                                                                                                                      |
| Bandpass    | $Z^{-1} = -\frac{z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + \frac{k-1}{k+1}}{\frac{k-1}{k+1} z^{-2} - \frac{2\alpha k}{k+1} z^{-1} + 1}$ | $\alpha = \frac{\cos\left(\frac{\omega_{p2} + \omega_{p1}}{2}\right)}{\cos\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right)}$ $k = \cot\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right) \tan\left(\frac{\theta_p}{2}\right)$ $\omega_{p1} = \text{desired lower cutoff freq.}$ $\omega_{p2} = \text{desired upper cutoff freq.}$ |



|          |                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                      |
|----------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Bandstop | $Z^{-1} = \frac{z^{-2} - \frac{2\alpha}{1+k}z^{-1} + \frac{1-k}{1+k}}{\frac{1-k}{1+k}z^{-2} - \frac{2\alpha}{1+k}z^{-1} + 1}$ | $\alpha = \frac{\cos\left(\frac{\omega_{p2} + \omega_{p1}}{2}\right)}{\cos\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right)}$ $k = \tan\left(\frac{\omega_{p2} - \omega_{p1}}{2}\right)\tan\left(\frac{\theta_p}{2}\right)$ <p><math>\omega_{p1}</math> = desired lower cutoff freq.</p> <p><math>\omega_{p2}</math> = desired upper cutoff freq.</p> |
|----------|-------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## ✧ Four Types of Linear FIR Filters

- Type 1:  $h[n]=h[M-n]$ ,  $0 \leq n \leq M$ , with  $M$  is an even integer.

$$\begin{aligned}
 H(e^{j\omega}) &= \sum_{n=0}^M h[n]e^{-j\omega n} \\
 &= h[M/2]e^{-j\omega M/2} + \sum_{n=0}^{M/2} \{h[n]e^{-j\omega n} + h[M-n]e^{-j\omega(M-n)}\} \\
 &= h[M/2]e^{-j\omega M/2} + \sum_{n=0}^{M/2-1} h[n]e^{-j\omega M/2} [e^{-j\omega(n-M/2)} + e^{j\omega(n-M/2)}] \\
 &= e^{-j\omega M/2} \{h[M/2] + \sum_{n=0}^{M/2-1} 2h[n]\cos(\omega(n-M/2))\}
 \end{aligned}$$

- Let  $a[0]=h[M/2]$ ,  $a[k]=2h[M/2-k]$ ,  $k=1, 2, \dots, M/2$ .

- The delay  $M/2$  is an integer.

$$H(e^{j\omega}) = e^{-j\omega M/2} \sum_{n=0}^{M/2} a[n]\cos(\omega n)$$

- Type 2:  $h[n]=h[M-n]$ ,  $0 \leq n \leq M$ , with  $M$  is an odd integer.

- Let  $b[k]=2h[(M+1)/2-k]$ ,  $k=1, 2, \dots, (M+1)/2$ .

- The delay  $M/2$  is an integer plus one-half.

$$H(e^{j\omega}) = e^{-j\omega M/2} \sum_{n=1}^{(M+1)/2} b[n]\cos[\omega(n-1/2)]$$

- Type 3:  $h[n]=-h[M-n]$ ,  $0 \leq n \leq M$ , with  $M$  is an even integer.

- Let  $c[k]=2h[M/2-k]$ ,  $k=1, 2, \dots, M/2$ .

- The delay  $M/2$  is an integer.

$$H(e^{j\omega}) = je^{-j\omega M/2} \sum_{n=1}^{M/2} c[n] \sin(\omega n)$$

- Type 4:  $h[n] = -h[M-n]$ ,  $0 \leq n \leq M$ , with  $M$  is an odd integer.

■ Let  $d[k] = 2h[(M+1)/2 - k]$ ,  $k = 1, 2, \dots, (M+1)/2$ .

■ The delay  $M/2$  is an integer plus one-half.

$$H(e^{j\omega}) = je^{-j\omega M/2} \sum_{n=1}^{(M+1)/2} d[n] \sin[\omega(n - 1/2)]$$

### ✧ Zero Phase System

- Type 1:  $h[n] = h[-n]$ ,  $-M/2 \leq n \leq M/2$ ,  $M$ : even integer.
- Type 3:  $h[n] = -h[-n]$ ,  $-M/2 \leq n \leq M/2$ ,  $M$ : even integer.

### ✧ Design of FIR Filters by Windowing

- Why FIR filters?
  - Always stable
  - *Exact* linear phase
  - Less sensitive to inaccurate coefficients
- <Disadvantage> Higher complexity (storage, multiplication) due to higher orders
- Design Methods
  - (1) Windowing
  - (2) Frequency sampling
  - (3) Computer-aided design

*Remark:* No meaningful analog FIR filters

- Windowing technique advantages
  - Simple
  - Pick up a “segment” (window) of the ideal (infinite)  $h_d[n]$
  - Filter order = window length =  $(M+1)$

General form:  $h[n] = h_d[n]w[n]$

Filter impulse response = Desired response x Window

**Example: Rectangular window**

$$\text{Window shape: } w[n] = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

$$\rightarrow h[n] = \begin{cases} h_d[n], & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

- Because the filter specifications are (often) given in the frequency domain  $H_d(e^{j\omega})$ . We take the inverse DTFT to obtain  $h_d[n]$ .

$$h_d[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\omega}) e^{j\omega n} d\omega$$

$$\text{Or, } H_d(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h_d[n] e^{-j\omega n}$$

Now, because of the inclusion of  $w[n]$ ,

$$H(e^{j\omega}) = \frac{1}{2\pi} \int_{-\pi}^{\pi} H_d(e^{j\theta}) \cdot W(e^{j(\omega-\theta)}) d\theta \quad (\text{A periodic convolution})$$

That is,  $H(e^{j\omega})$  is “smeared” version of  $H_d(e^{j\omega})$ . (Fig.7.27, P.564)

Why  $W(e^{j\omega})$  cannot be  $\delta(e^{j\omega})$ ? (If so,  $H(e^{j\omega}) = H_d(e^{j\omega})$ !)

**Parameters** (to choose): (1) Window size (order of filter)  
(2) Window shape

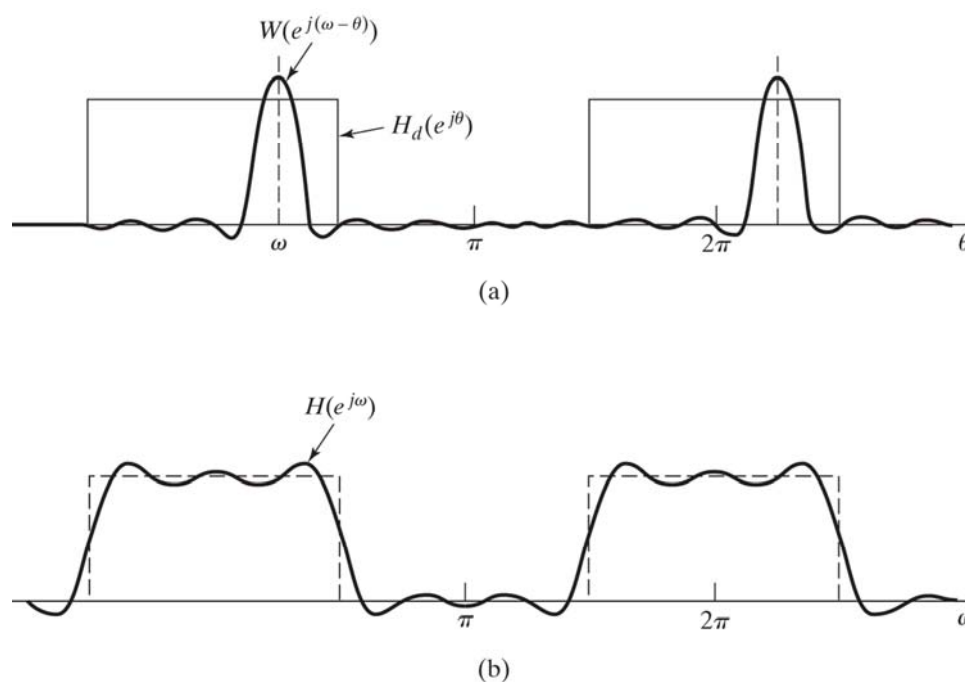


Figure 7.27 .

- **Rectangular Window:**  $w[n] = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$

- Narrow mainlobe
- High sidelobe (Gibbs phenomenon)
- Frequency response (Fig.7.20, P.468)

$$W(e^{j\omega}) = \sum_{n=0}^M 1 \cdot e^{-j\omega n} = \frac{1 - e^{-j\omega(M+1)}}{1 - e^{-j\omega}} = e^{-j\omega \frac{M}{2}} \frac{\sin\left[\omega \frac{(M+1)}{2}\right]}{\sin\left(\frac{\omega}{2}\right)}$$

- Mainlobe  $\sim \frac{4\pi}{M+1}$ ,  $M \uparrow$ ,  $W(e^{j\omega}) \rightarrow \delta(e^{j\omega})$

- Peak sidelobe  $\sim -13$  dB (lower than the mainlobe)

Area under each lobe remains constant with increasing  $M$

→ Increasing  $M$  does not lower the (relative) amplitude of the sidelobe. (Gibbs phenomenon)

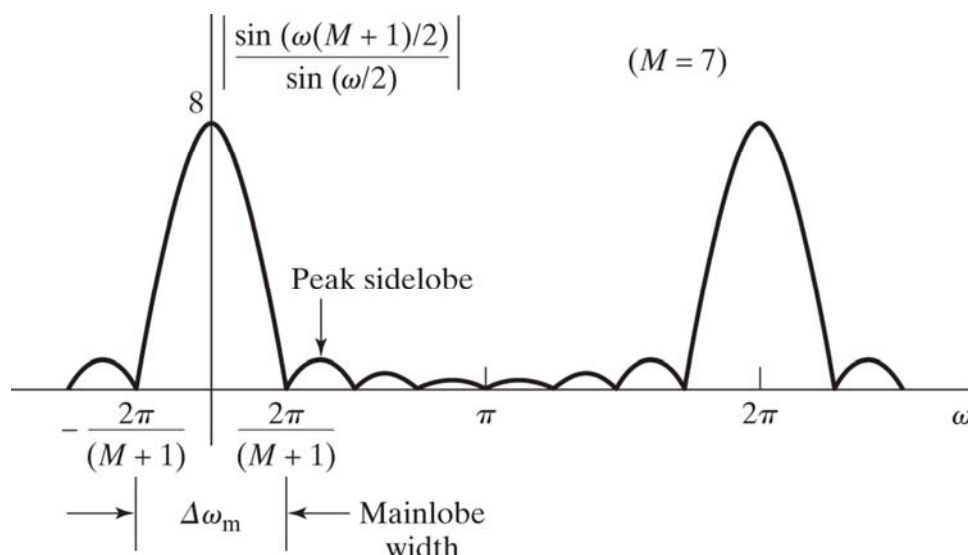


Fig. 7.28

*Remarks:* For frequency selective filters (ideal lowpass, high-pass, ...), narrow mainlobe  $\rightarrow$  sharp transition  
 lower sidelobe  $\rightarrow$  oscillation reduction  
 Hence, other windows were invented.

- Commonly Used Windows (Fig.7.29, P.565)
  - Sidelobe amplitude (area) vs. mainlobe width
  - Closed form, easy to compute

### **Bartlett (triangular) Window:**

$$w[n] = \begin{cases} \frac{2n}{M}, & 0 \leq n \leq \frac{M}{2} \\ 2 - \frac{2n}{M}, & \frac{M}{2} < n \leq M \\ 0, & \text{otherwise} \end{cases}$$

### **Hanning Window:**

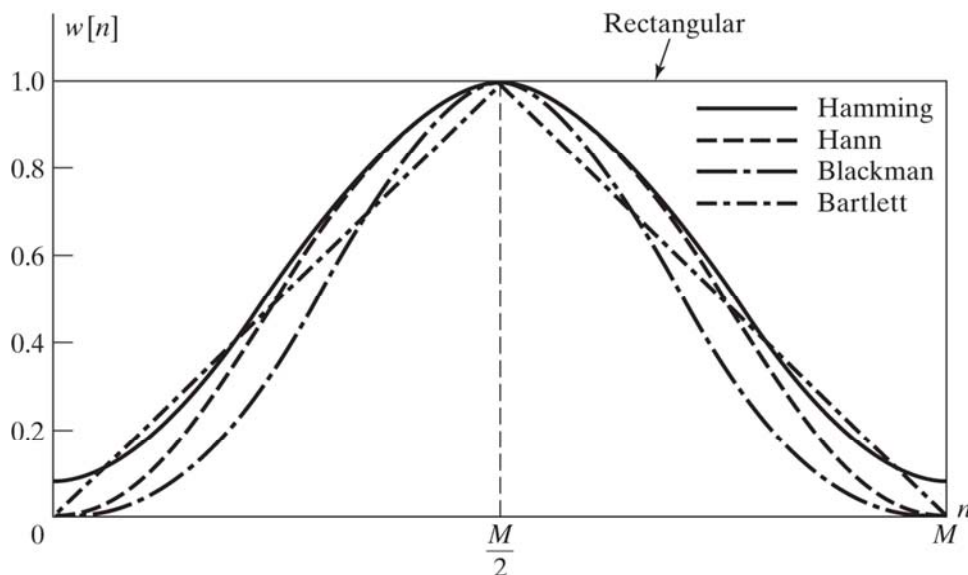
$$w[n] = \begin{cases} 0.5 - 0.5 \cos\left(\frac{2n}{M} \pi\right), & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

**Hamming Window:**

$$w[n] = \begin{cases} 0.54 - 0.46 \cos\left(\frac{2n}{M} \pi\right), & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

**Blackman Window:**

$$w[n] = \begin{cases} 0.42 - 0.5 \cos\left(\frac{2n}{M} \pi\right) + 0.08 \cos\left(\frac{4n}{M} \pi\right), & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

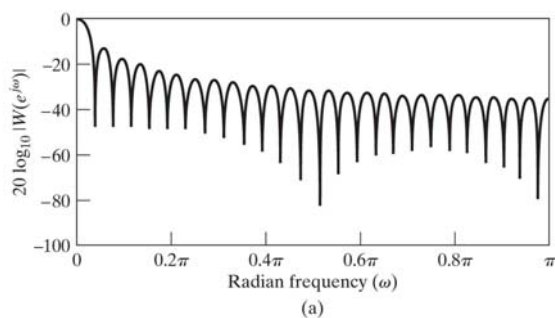


- *Comparison of Commonly Used Windows*

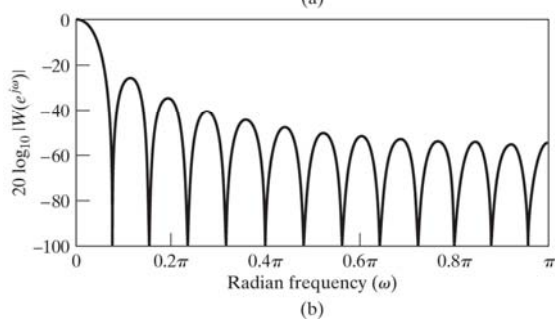
| Window Type | Peak Side-lobe Amplitude (Relative) | Approximate Width of Mainlobe | Equivalent Kaiser Window $\beta$ | Transition Width of Equivalent Kaiser Window |
|-------------|-------------------------------------|-------------------------------|----------------------------------|----------------------------------------------|
| Rectangular | -13                                 | $4\pi/(M+1)$                  | 0                                | $1.81\pi/M$                                  |
| Bartlett    | -25                                 | $8\pi/M$                      | 1.33                             | $2.37\pi/M$                                  |
| Hanning     | -31                                 | $8\pi/M$                      | 3.86                             | $5.01\pi/M$                                  |
| Hamming     | -41                                 | $8\pi/M$                      | 4.86                             | $6.27\pi/M$                                  |
| Blackman    | -57                                 | $12\pi/M$                     | 7.04                             | $9.19\pi/M$                                  |

Frequency response of commonly used windows (Fig.7.30, pp.566-567)

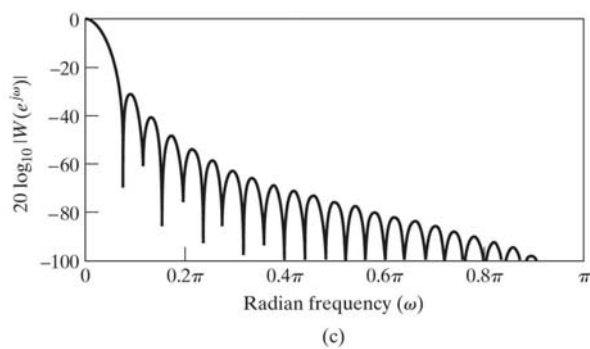
Rectangular



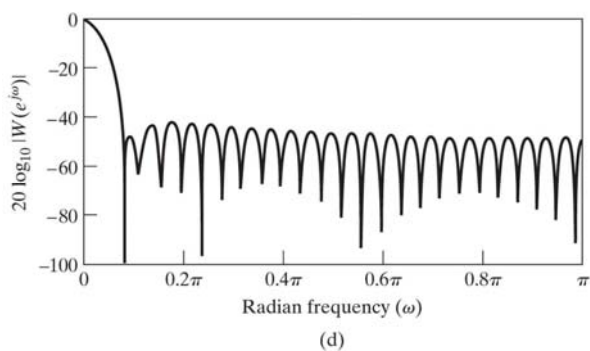
Bartlett



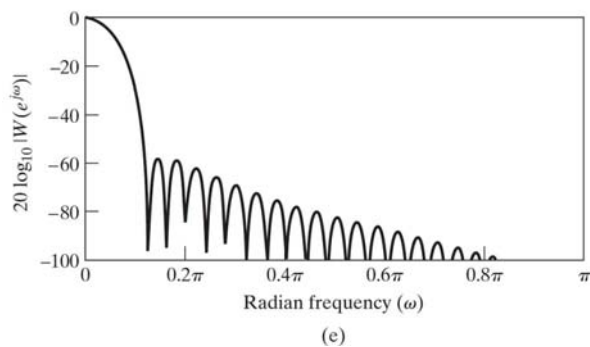
Hann



Hamming



Blackman





- Generalized Linear Phase Filters

-- We wish  $H(e^{j\omega})$  be (general) linear phase.

<Window> Choose windows such that

$$w[n] = w[M - n], \quad 0 \leq n \leq M$$

That is, symmetric about  $M/2$  (samples)

$$W(e^{j\omega}) = W_e(e^{j\omega}) \cdot e^{-j\omega \frac{M}{2}}, \text{ where } W_e(e^{j\omega}) \text{ is real.}$$

<Desired filter> Suppose the desired filter is also generalized linear phase

$$H_d(e^{j\omega}) = H_e(e^{j\omega}) \cdot e^{-j\omega \frac{M}{2}}$$

<Filter>  $H(e^{j\omega})$  is a periodic convolution of  $H_d(e^{j\omega})$  and

$$W(e^{j\omega})$$

$$\begin{aligned} H(e^{j\omega}) &= \frac{1}{2\pi} \int_{-\pi}^{\pi} H_e(e^{j\theta}) \cdot W_e(e^{j(\omega-\theta)}) \cdot e^{-j\theta \frac{M}{2}} e^{-j\frac{(\omega-\theta)M}{2}} d\theta \\ &= \frac{1}{2\pi} \underbrace{\int_{-\pi}^{\pi} H_e(e^{j\theta}) \cdot W_e(e^{j(\omega-\theta)}) d\theta}_{A_e(e^{j\omega})} \cdot e^{-j\omega \frac{M}{2}} \end{aligned}$$

$$A_e(e^{j\omega}) \text{ is real.}$$

Thus,  $H(e^{j\omega})$  is also generalized linear phase.

*Example:* Linear phase lowpass filter (P.569)

$$\text{Ideal lowpass: } H_{lp}(e^{j\omega}) = \begin{cases} e^{-j\omega \frac{M}{2}}, & |\omega| < \omega_c \\ 0, & \omega_c < |\omega| \leq \pi \end{cases}$$

$$\text{Impulse response: } h_{lp}[n] = \frac{\sin\left[\omega_c\left(n - \frac{M}{2}\right)\right]}{\pi\left(n - \frac{M}{2}\right)}$$

$$\text{Designed filter: } h[n] = \frac{\sin\left[\omega_c\left(n - \frac{M}{2}\right)\right]}{\pi\left(n - \frac{M}{2}\right)} \cdot w[n]$$

$\omega_c$ : 1/2 amplitude of  $H(e^{j\omega})$  = cutoff frequency of the ideal lowpass filter

Peak to the left of  $\omega_c$  occurs at  $\sim 1/2$  mainlobe width

-Peak to the right of  $\omega_c$  occurs at  $\sim 1/2$  mainlobe width

Transition bandwidth  $\Delta\omega \sim$  mainlobe width- (smaller)

Peak approximation error: proportional to sidelobe area

(Fig. 7.31, P.570)

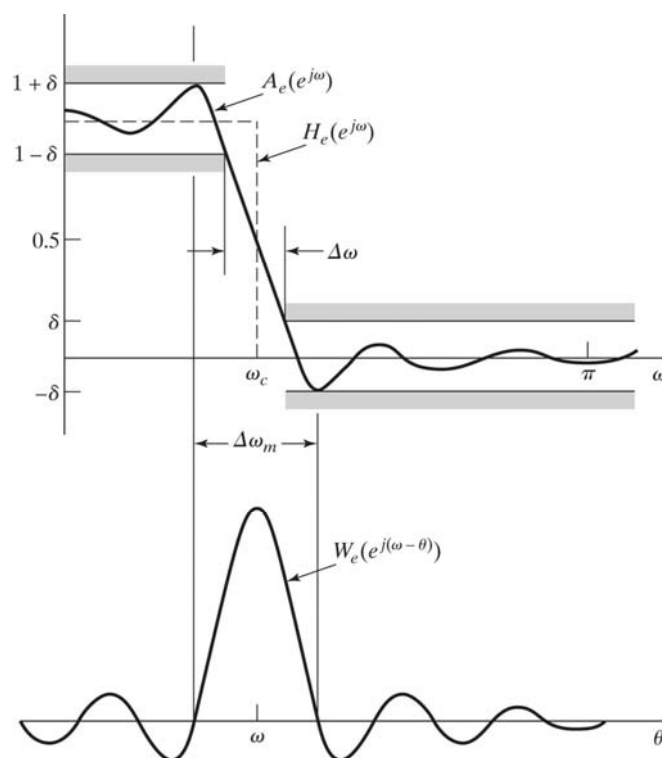


Figure 7.31 Illustration of type of approximation obtained at a discontinuity of the ideal frequency response.

- **Kaiser Window**

-- Nearly optimal trade-off between mainlobe width and sidelobe area

$$w[n] = \begin{cases} \frac{I_0 \left[ \beta \left( 1 - \left[ \frac{(n - \alpha)}{\alpha} \right]^2 \right)^{1/2} \right]}{I_0(\beta)}, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$$

where  $I_0(\cdot)$ : zeroth-order modified Bessel function of the first kind

$\alpha: M/2$

$\beta$ : shape parameter;  $\beta = 0$ , rectangular window

$\beta \uparrow$ , mainlobe width  $\uparrow$ , sidelobe area  $\downarrow$

(Fig. 7.32, P.572)

How to select  $\alpha, \beta$ ?

Based on approximation error  $\delta$  and transition bandwidth

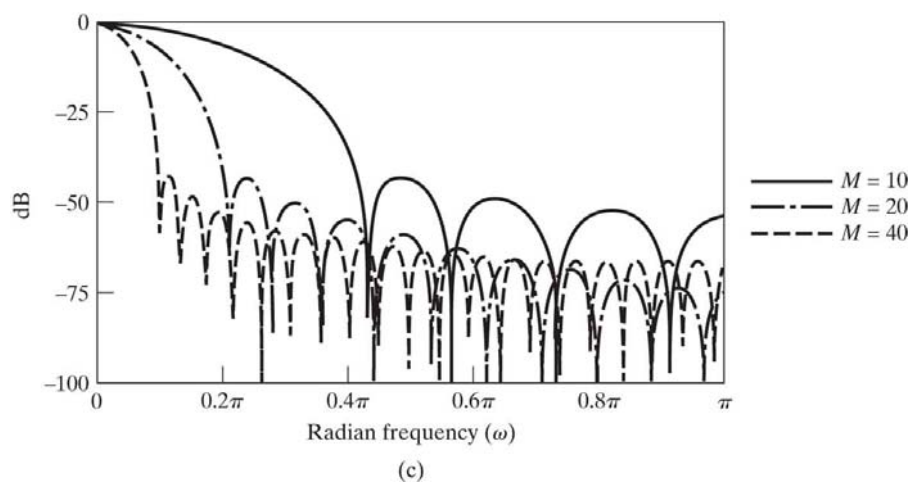
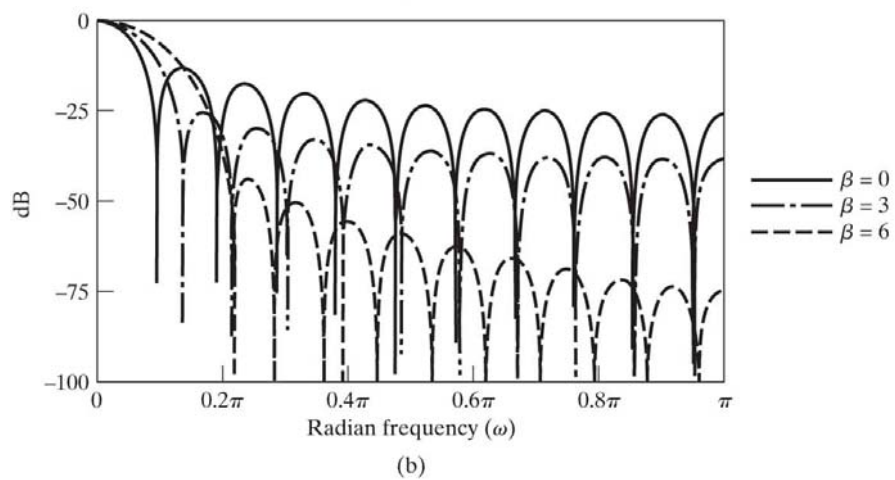
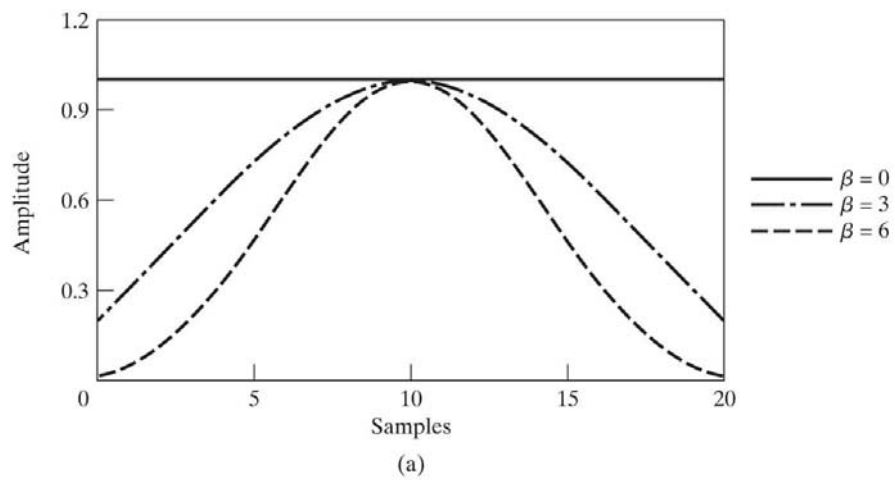
$$\Delta\omega$$

$$-- A \equiv -20 \cdot \log_{10} \delta$$

$$\beta = \begin{cases} 0.1102(A - 8.7), & A > 50 \\ 0.5842(A - 21)^{0.4} + 0.07886(A - 21), & 21 \leq A \leq 50 \\ 0.0 & A < 21 \end{cases}$$

$$-- \Delta\omega = \omega_s - \omega_p \text{ (stopband - passband)}$$

$$M = \frac{A - 8}{2.285 \cdot \Delta\omega} \text{ (within } \pm 2 \text{ over a wide range of } \Delta\omega \text{ and } A)$$



- *Kaiser window example* – lowpass

Specifications:  $\delta_1 = \delta_2 = 0.001$

Ideal lowpass cutoff:  $\omega_c = \frac{\omega_s + \omega_p}{2} = 0.5\pi$

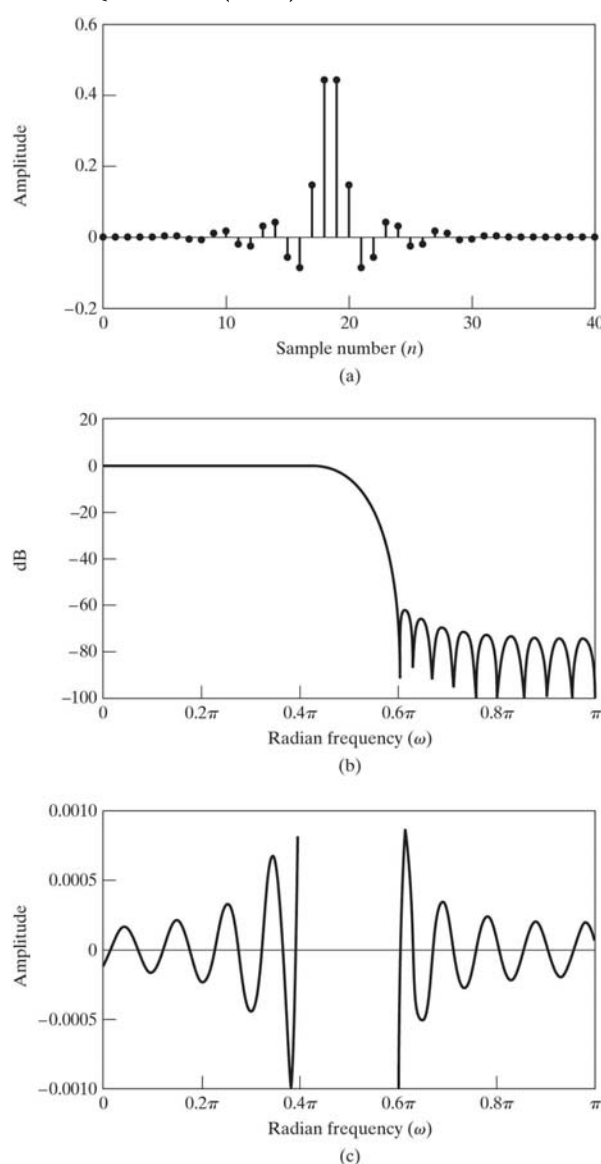
Select parameters:  $\begin{cases} \Delta\omega = \omega_s - \omega_p = 0.2\pi \\ A = -20\log_{10} \delta = 60 \end{cases} \rightarrow \begin{cases} \beta = 5.653 \\ M = 37 \end{cases}$

$$\alpha = M/2 = 18.5$$

This is a type II, linear phase (odd M, even symmetry) filter.

Approximation error:  $|H_d(e^{j\omega})| - |H(e^{j\omega})|$

$$E_A(e^{j\omega}) = \begin{cases} 1 - A_e(e^{j\omega}), & 0 \leq \omega < \omega_p \\ 0 - A_e(e^{j\omega}), & \omega_s < \omega \leq \pi \end{cases} \quad (\text{Fig.7.34, P.575})$$



**Figure 7.34** Response functions for the lowpass filter designed with a Kaiser window (a) Impulse response ( $M = 37$ ). (b) Log magnitude. (c) Approximation error for  $A_e(e^{j\omega})$

- *Kaiser window example* – highpass

$$\text{Ideal highpass: } H_{\text{hp}}(e^{j\omega}) = \begin{cases} 0, & 0 \leq |\omega| < \omega_c \\ e^{-j\omega \frac{M}{2}}, & \omega_c < |\omega| \leq \pi \end{cases}$$

$$h_{\text{hp}}[n] = \frac{\sin \pi \left( n - \frac{M}{2} \right)}{\pi \left( n - \frac{M}{2} \right)} - \frac{\sin \omega_c \left( n - \frac{M}{2} \right)}{\pi \left( n - \frac{M}{2} \right)}$$

Specifications:  $\delta_1 = \delta_2 = 0.021$

$$\text{Highpass cutoff: } \omega_c = \frac{\omega_s + \omega_p}{2} = \frac{0.35\pi + 0.5\pi}{2}$$

$$\text{Select parameters: } \begin{cases} \Delta\omega \\ A \end{cases} \rightarrow \begin{cases} \beta = 2.6 \\ M = 24 \end{cases}$$

This is a Type I filter.

*Check!* Approximation error = 0.0213 > 0.021!!

Increase M to 25 → Not good! Why? (Fig.7.36, P.578)

This is a Type II filter: a zero at  $-1$ . →  $H_d(e^{j\pi}) = 0$

But we want it to be 1 because this is a highpass filter.

Increase M to 26. Okay!

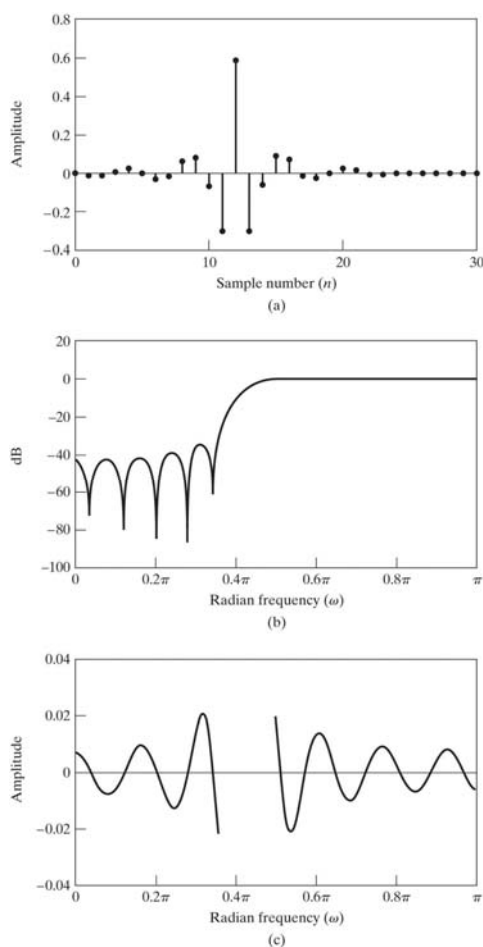


Fig. 7.35

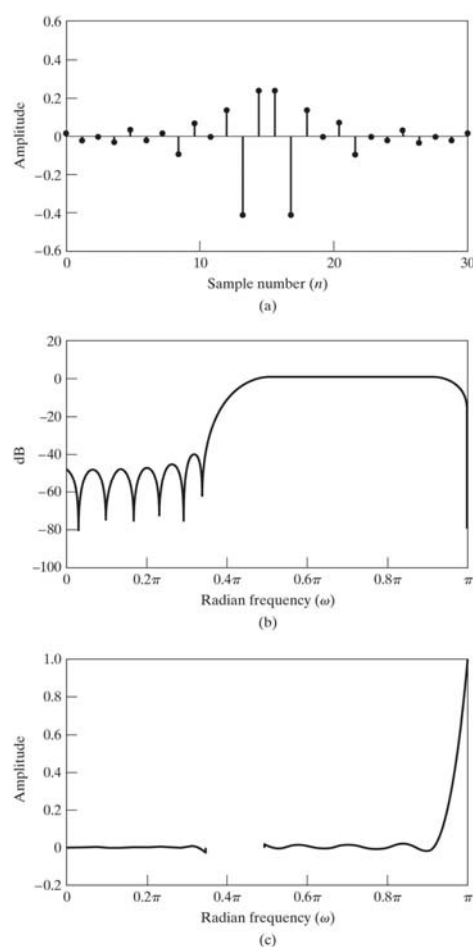


Fig. 7.36

- *Kaiser window example – differentiator*

Ideal differentiator:  $\sim \frac{d}{dt}$

$$H_{\text{diff}}(e^{j\omega}) = (j\omega) \cdot e^{-j\omega \frac{M}{2}}, \quad -\pi < \omega < \pi$$

$$h_{\text{diff}}[n] = \frac{\cos \pi \left( n - \frac{M}{2} \right)}{\left( n - \frac{M}{2} \right)} - \frac{\sin \pi \left( n - \frac{M}{2} \right)}{\pi \left( n - \frac{M}{2} \right)^2}$$

Note that both terms in  $h_{\text{diff}}[n]$  are odd symmetric. Hence,

$$h[n] = -h[M - n].$$

This must be a Type III or Type IV system.

## &lt;Comparison&gt;

Case 1:  $M=10$ ,  $\beta = 2.4 \rightarrow$  Type III (Fig. 7.38, P.581)

Zeros at 1 and  $-1$ . Approximation is not good at  $\omega = \pi$ .

Case 2:  $M=5$ ,  $\beta = 2.4 \rightarrow$  Type IV (Fig. 7.39, P.582)

Zeros at 1. Approximation error is smaller.

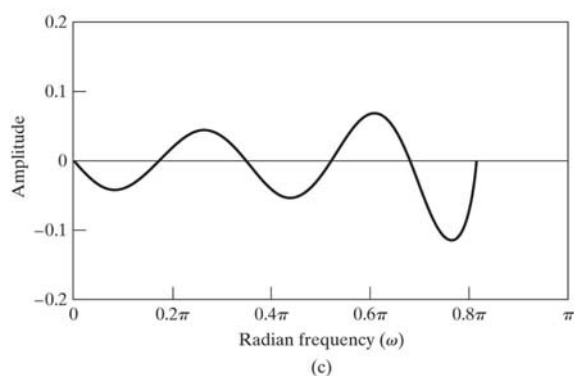
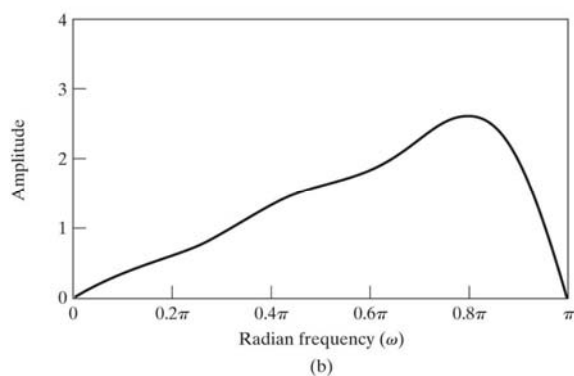
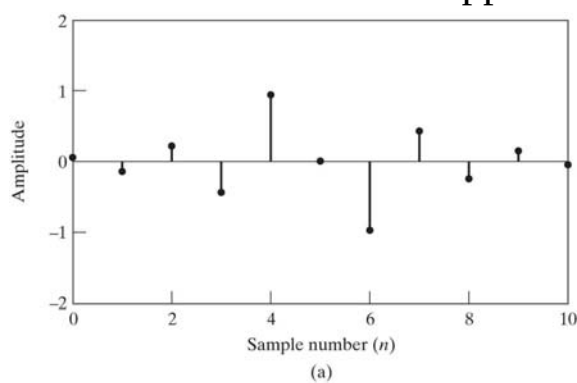


Fig. 7.38

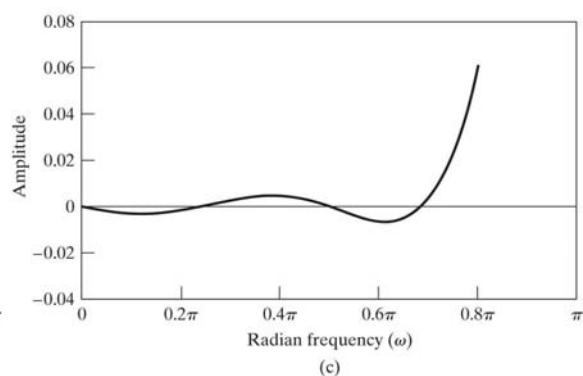
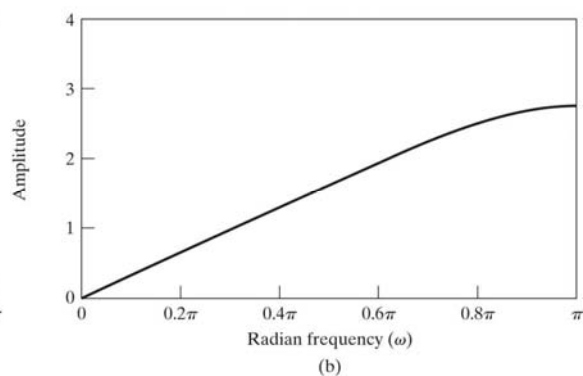
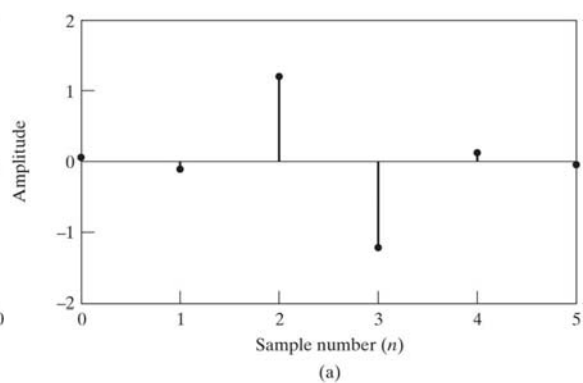


Fig. 7.39



## 5.4 Optimum Approximation of FIR Filters

- Why computer-aided design?
  - Optimum: minimize an error criterion
  - More freedom in selecting constraints.
 (In windowing method: must  $\delta_1 = \delta_2 = \delta$ )
- Several algorithms – **Parks-McClellan algorithm** (1972)

### Type I linear phase FIR filter

Its symmetry property:  $h_e[n] = h_e[-n]$  (omit delay)

Check its frequency response:

$$\begin{aligned}
 A_e(e^{j\omega}) &= \sum_{n=-L}^L h_e[n] \cdot e^{-j\omega n} \\
 &= h_e[0] + \sum_{n=1}^L 2h_e[n] \cdot \cos(\omega n) \\
 &= a_0 + \sum_{k=1}^L a_k \cdot (\cos(\omega))^k \\
 &= \sum_{k=0}^L a_k \cdot (\cos(\omega))^k \\
 &= P(x) \Big|_{x=\cos \omega}
 \end{aligned}$$

Note that  $P(x) = \sum a_k x^k$  is an  $L$ th-order polynomial. In the above process, we use a polynomial expression of  $\cos(\cdot)$ ,  $\cos(\omega n) = T_n(\cos \omega)$ , where  $T_n(\cdot)$  is the  $n$ th-order Chebyshev polynomial. Thus, we shift our goal from finding  $(L+1)$  values of  $\{h_e[n]\}$  to finding  $(L+1)$  values of  $\{a_k\}$ .

(Why are we doing this? We want to use the polynomial approximation algorithms.)

<Our Problem now>

Adjustable parameters:  $\{a_k\}$ ,  $(L+1)$  values

Specifications:  $\omega_p, \omega_s, \delta_1/\delta_2 = K$ , and  $L$  ( $L$  is often preselected)

Error criterion:  $E(\omega) = W(\omega) \cdot [H_d(e^{j\omega}) - A_e(e^{j\omega})]$

Goal: minimize the maximum error

$$\min_{\{h_e[n]\}^L} \left( \max_{\omega \in F} |E(\omega)| \right), \quad F: \text{passband and stopband}$$

(Note: Often, no constraint on the transition band)

(Why choose this minimization target? Even error values!

Recall: In the rectangular windowing method, we actually mi-

nimize  $\varepsilon^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |H_d(e^{j\omega}) - H(e^{j\omega})|^2 d\omega$ . Although the total

squared error can be small but errors at some frequencies may be large.)

**<Alternation Theorem>** (Figs.7.44, 7.45; pp.589, 590)

$F_p$ : closed subset consists of (the union) of disjoint closed subsets of the real axis  $x$  *Example, lowpass:*  
 $[0, \omega_p], [\omega_s, \pi]$   
 $\rightarrow x = \cos \omega \rightarrow$   
 $[1, \cos \omega_p], [\cos \omega_s, 1]$

$P(x)$ :  $r$ th-order polynomial

$$P(x) = \sum_{k=0}^r a_k x^k$$

$$P(\cos \omega) = \sum_{k=0}^L a_k (\cos \omega)^k$$

$D_p(x)$ : desired function of  $x$  continuous on  $F_p$

$$D_p(x) = \begin{cases} 1, & x_p \leq x \leq 1 \\ 0, & -1 \leq x \leq x_s \end{cases}$$

$$x = \cos \omega$$

$W_p(x)$ : weighting: positive, continuous on  $F_p$

$$W_p(x) = \begin{cases} 1/K, & x_p \leq x \leq 1 \\ 1, & -1 \leq x \leq x_s \end{cases}$$

$E_p(x)$ : weighted error

$$E_P(x) = W_P(x)[D_P(x) - P(x)] \quad E_P(x) = W_P(x)[D_P(x) - P(x)]$$

$\|E\|$ : maximum error

$$\|E\| = \max_{x \in F_P} E_P(x) \quad \|E\| = \delta_2$$

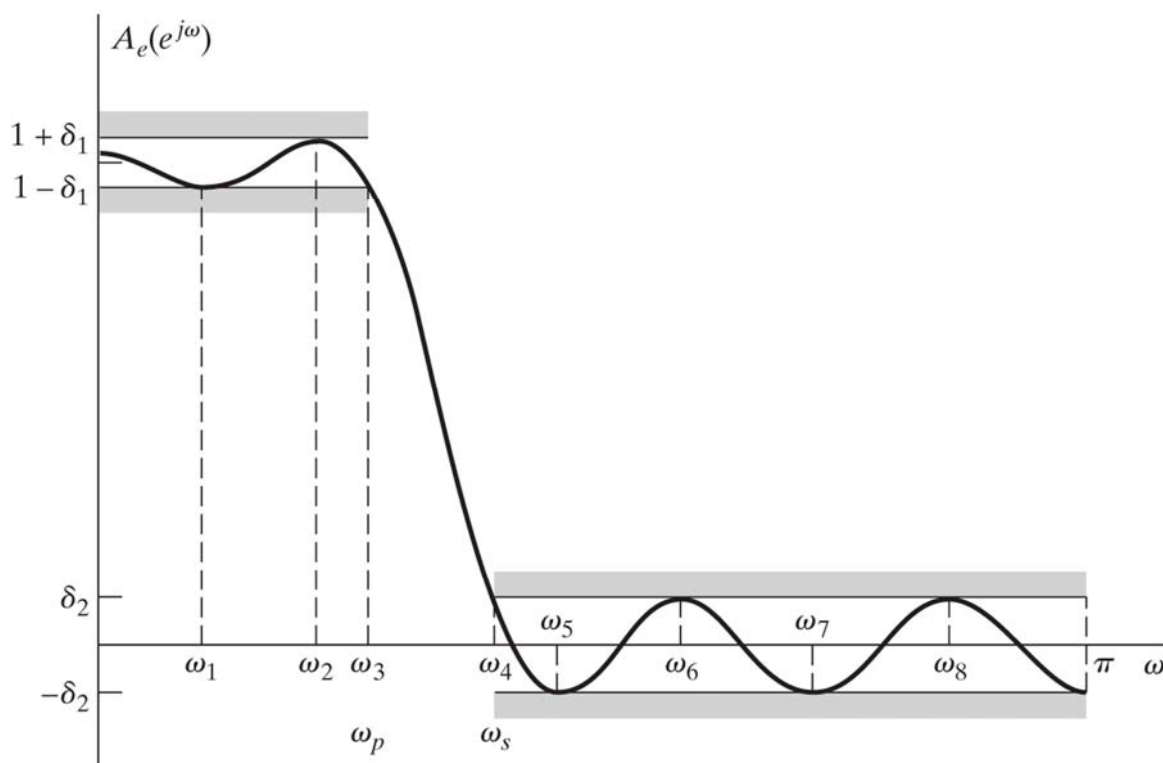
$P(x)$  is the *unique*  $r$ th-order polynomial that minimizes  $\|E\|$   
 if and only if  $E_P(x)$  exhibits *at least*  $(r+2)$  **alternations**

**Alternation:** There exist  $(r+2)$  values  $x_i$  in  $F_P$  such that

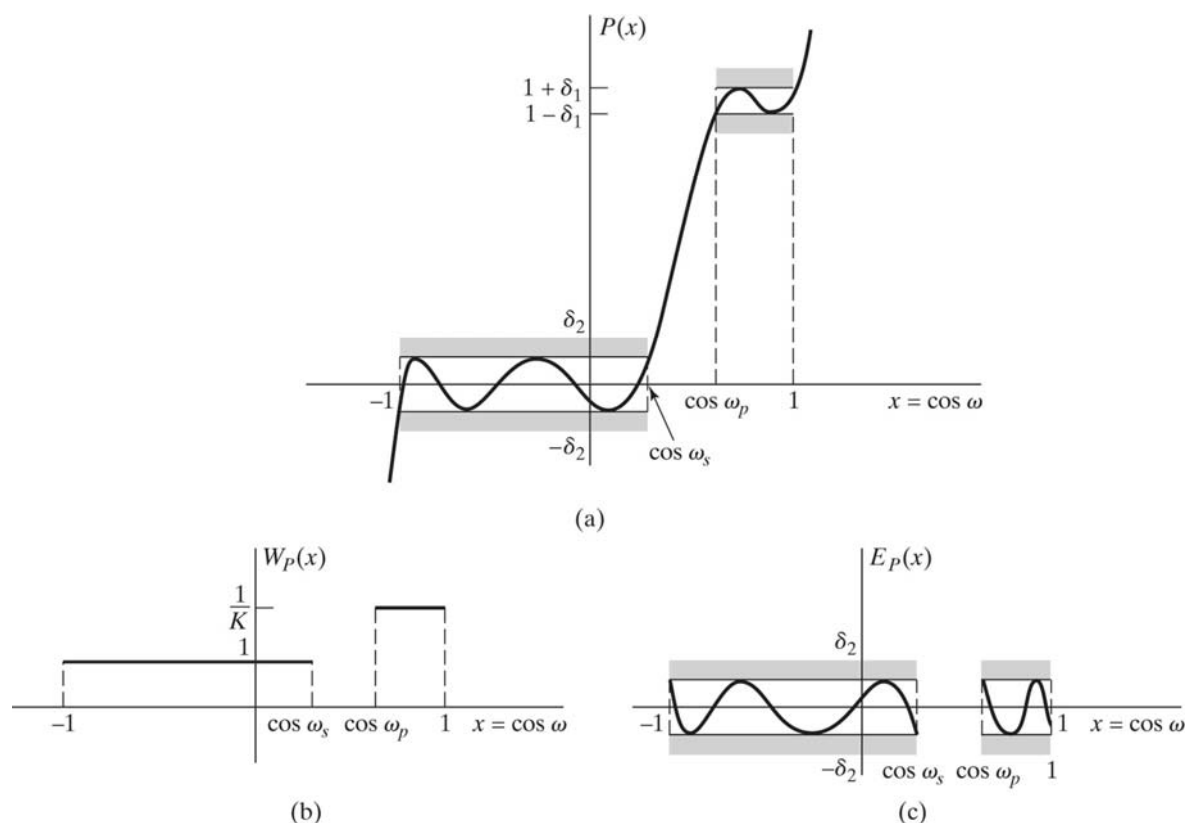
$$E_P(x_i) = -E_P(x_{i+1}) = \pm\|E\|, i = 1, 2, \dots, (r+1), \text{ where}$$

$$x_1 < x_2 < \dots < x_{r+2}.$$

*Remark:* Two conditions here for alternation: value and sign.



**Figure 7.44** typical example of a lowpass filter approximation that is optimal according to the alternation theorem for  $l = 7$



**Figure 7.45** Equivalent polynomial approximation functions as a function of  $x = \cos \omega$ . (a) Approximating polynomial. (b) Weighting function. (c) Approximation error.

### <Type I linear phase FIR filter>

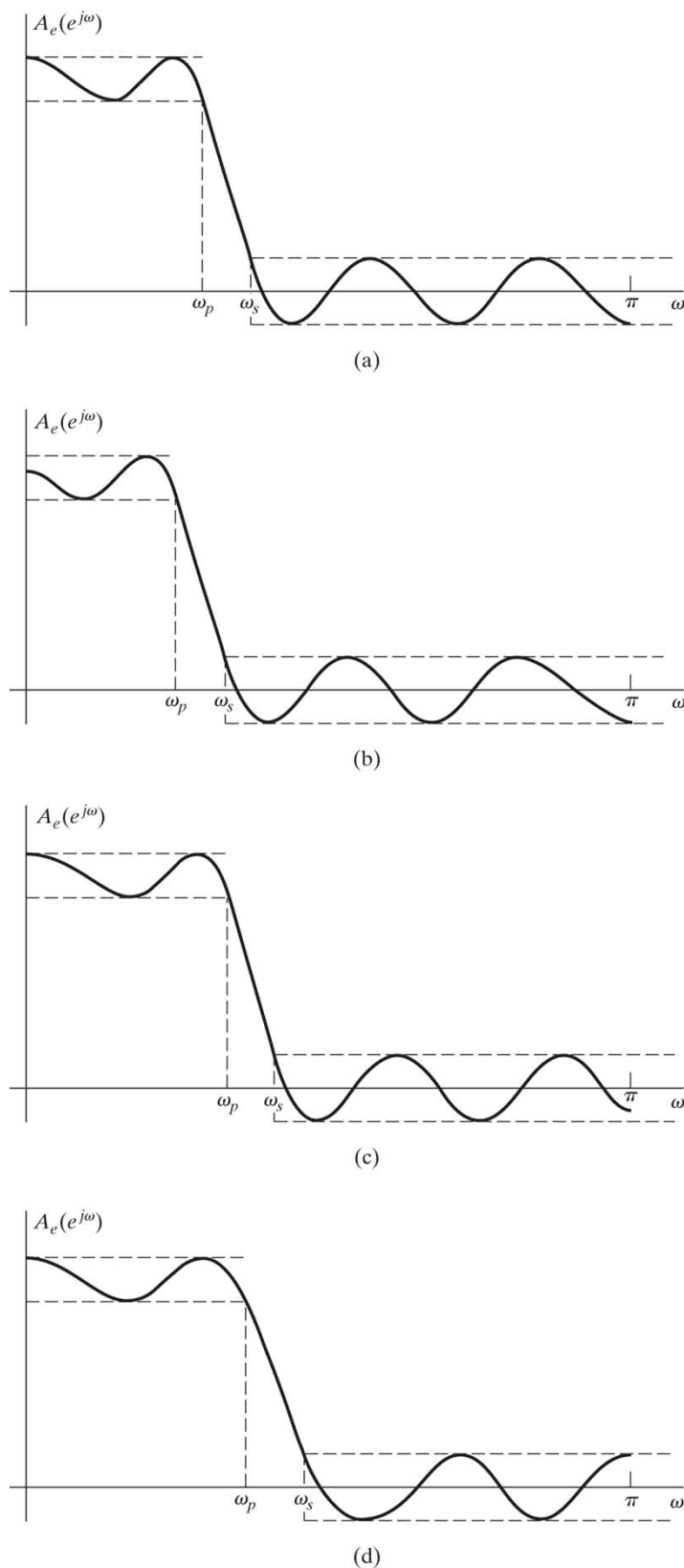
- (1) Maximum number of alternations of errors =  $(L+3)$
  - (2) Alternations always occur at  $\omega_p$  and  $\omega_s$
  - (3) Equiripple except possibly at  $\omega = 0$  and  $\omega = \pi$
- (Reasons) (O&S 3<sup>rd</sup>, pp.590-593)

(a) Locations of extrema:  $L$ th-order polynomial has at most  $L-1$  extrema. Now, in addition, the local extrema may locate at band edges  $\omega = 0, \pi, \omega_p, \omega_s$ . Hence, at most, there are  $(L+3)$  extrema or alternations.

(Note: Because  $x = \cos \omega$ ,  $\frac{dP(\cos \omega)}{d\omega} = 0$ , at  $\omega = 0$  and

$\omega = \pi$ .)

(Fig.7.46, P.591 – 4 different cases)



**Figure 7.46** Possible optimum lowpass filter approximations for  $L = 7$ . (a)  $L + 3$  alternations (extranipple case). (b)  $L + 2$  alternations (extremum at  $\omega = \pi$ ). (c)  $L + 2$  alternations (extremum at  $\omega = 0$ ). (d)  $L + 2$  alternations (extremum at both  $\omega = 0$  and  $\omega = \pi$ ).

- (b) If  $\omega_p$  is not an alternation, for example, then because of the  $\pm$  sign sequence, we lose two alternations  $\rightarrow (L+1)$  alternations  $\leftarrow$  violates the  $(L+2)$  alternation theorem.

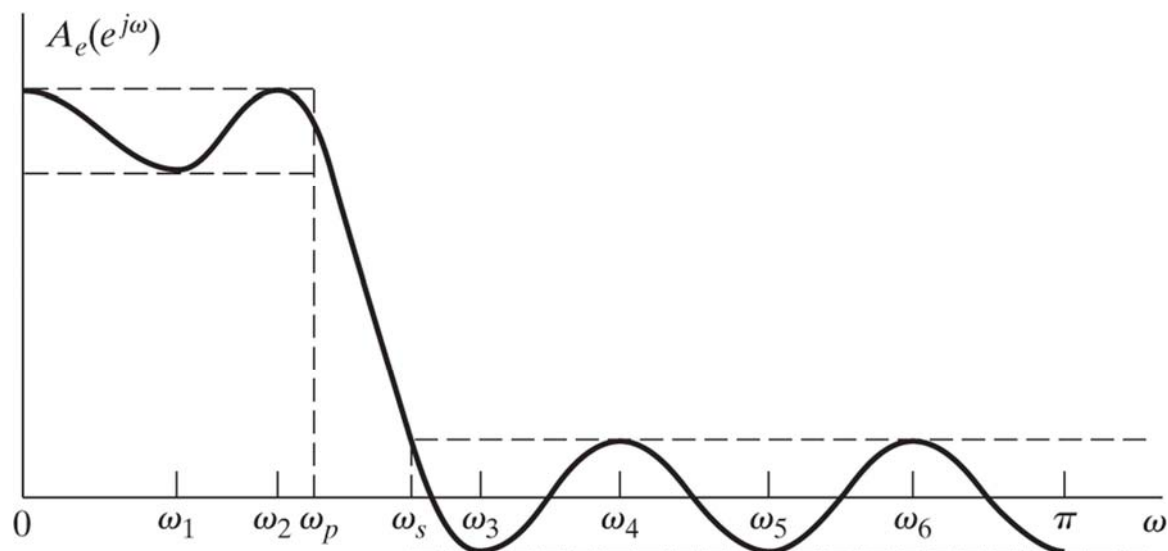


Figure 7.47 Illustration that the passband edge  $\omega_p$  must be an alternation frequency.

- (c) The only possibility that the extrema can be a non-alternation is that it locates at  $\omega = 0$  or  $\omega = \pi$ . In either case, we have  $(L+2)$  alternations – minimum requirement.

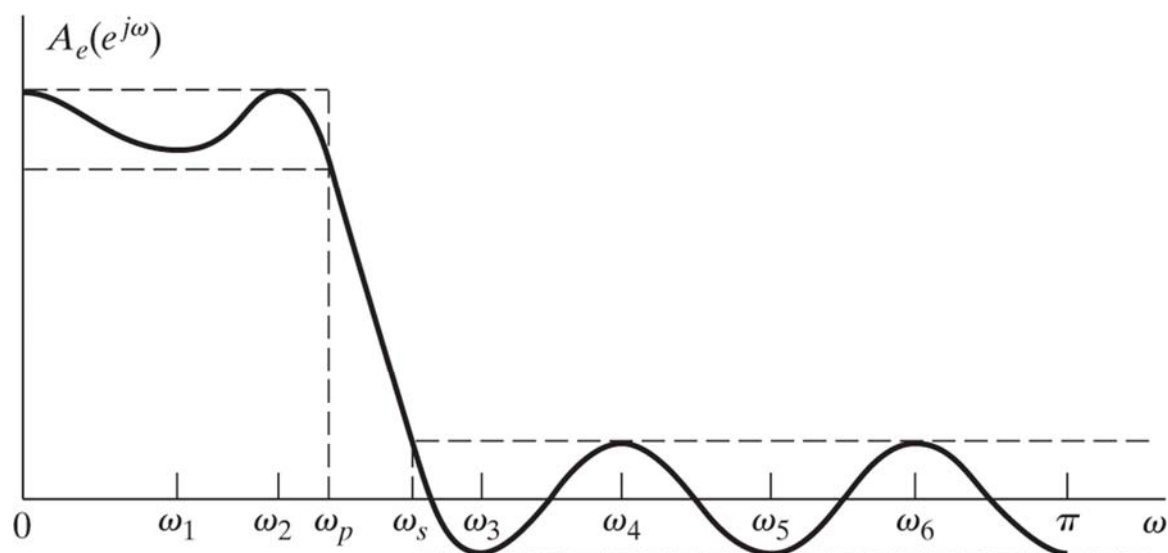


Figure 7.48 Illustration that the frequency response must be equiripple in the approximation bands.

## Type II linear phase FIR filter

Its symmetry property:  $h_e[n] = h_e[M - n]$ ,  $M$  odd

Frequency response:

$$\begin{aligned}
 H(e^{j\omega}) &= e^{-j\omega \frac{M}{2}} \left\{ \sum_{n=1}^{(M+1)/2} b[n] \cdot \cos(\omega(n - 1/2)) \right\} \\
 &= e^{-j\omega \frac{M}{2}} \cos\left(\frac{\omega}{2}\right) \left\{ \sum_{n=1}^{(M+1)/2} \tilde{b}[n] \cdot \cos(\omega n) \right\} \\
 \rightarrow H(e^{j\omega}) &= e^{-j\omega \frac{M}{2}} \cos\left(\frac{\omega}{2}\right) P(\cos \omega),
 \end{aligned}$$

$$\text{where } P(\cos \omega) = \sum_{k=0}^L a_k (\cos \omega)^k$$

*Problem:* How to handle  $\cos\left(\frac{\omega}{2}\right)$ ?

Transfer specifications!

$$\text{Let } H_d(e^{j\omega}) = D_p(\cos \omega) = \begin{cases} \frac{1}{\cos\left(\frac{\omega}{2}\right)}, & 0 \leq \omega \leq \omega_p \\ 0, & \omega_s \leq \omega \leq \pi \end{cases}$$

| Original                                                                            | New                                                                                         |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Ideal: $D(\cos \omega) \leftarrow \cos\left(\frac{\omega}{2}\right) P(\cos \omega)$ | Ideal: $\frac{D(\cos \omega)}{\cos\left(\frac{\omega}{2}\right)} \leftarrow P(\cos \omega)$ |

Thus,

$$W(\omega) = W_p(\cos \omega) = \begin{cases} \cos\left(\frac{\omega}{2}\right), & 0 \leq \omega \leq \omega_p \\ K, & \omega_s \leq \omega \leq \pi \end{cases}$$

- **Parks-McClellan Algorithm**

<Type I Lowpass>

According to the preceding theorems, errors

$E(\omega) = W(\omega) \cdot [H_d(e^{j\omega}) - A_e(e^{j\omega})]$  has alternations at  $\omega_i, i = 1, \dots, L + 2$ , if  $A_e(e^{j\omega})$  is the *optimum* solution.

That is, let  $\delta = \|E\|$ , the maximum error,

$$W(\omega_i) \cdot [H_d(e^{j\omega_i}) - A_e(e^{j\omega_i})] = (-1)^{i+1} \delta, \quad i = 1, 2, \dots, L + 2.$$

Because

$$A_e(e^{j\omega}) = \sum_{k=0}^L a_k (\cos \omega)^k = a_0 1 + a_1 \cos \omega + a_2 (\cos \omega)^2 + \dots,$$

$$\text{at } \omega_1: a_0 1 + a_1 \cos \omega_1 + a_2 (\cos \omega_1)^2 + \dots \leftrightarrow a_0 1 + a_1 x_1 + a_2 (x_1)^2 + \dots$$

$$\text{at } \omega_2: a_0 1 + a_1 \cos \omega_2 + a_2 (\cos \omega_2)^2 + \dots \leftrightarrow a_0 1 + a_1 x_2 + a_2 (x_2)^2 + \dots$$

...

Hence,

$$\begin{bmatrix} 1 & x_1 & x_1^2 & \dots & x_1^L & \frac{1}{W(\omega_1)} \\ 1 & x_2 & x_2^2 & \dots & x_2^L & \frac{-1}{W(\omega_2)} \\ \vdots & \ddots & & & & \\ 1 & x_{L+2} & x_{L+2}^2 & \dots & x_{L+2}^L & \frac{(-1)^{L+2}}{W(\omega_{L+2})} \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ \delta \end{bmatrix} = \begin{bmatrix} H_d(e^{j\omega_1}) \\ H_d(e^{j\omega_2}) \\ \vdots \\ H_d(e^{j\omega_{L+2}}) \end{bmatrix}$$

**Remark:** For Type I lowpass filter,  $\omega_p$  and  $\omega_s$  must be two of the alternation frequencies  $\{\omega_i\}$ .



Now, we have  $L+2$  simultaneous equations and  $L+2$  unknowns,  $\{a_i\}$  and  $\delta$ . The solutions are

$$\delta = \frac{\sum_{k=1}^{L+2} b_k H_d(e^{j\omega_k})}{\sum_{k=1}^{L+2} \frac{b_k (-1)^{k+1}}{W(\omega_k)}}, \quad b_k = \prod_{\substack{i=1 \\ i \neq k}}^{L+2} \frac{1}{(x_k - x_i)}.$$

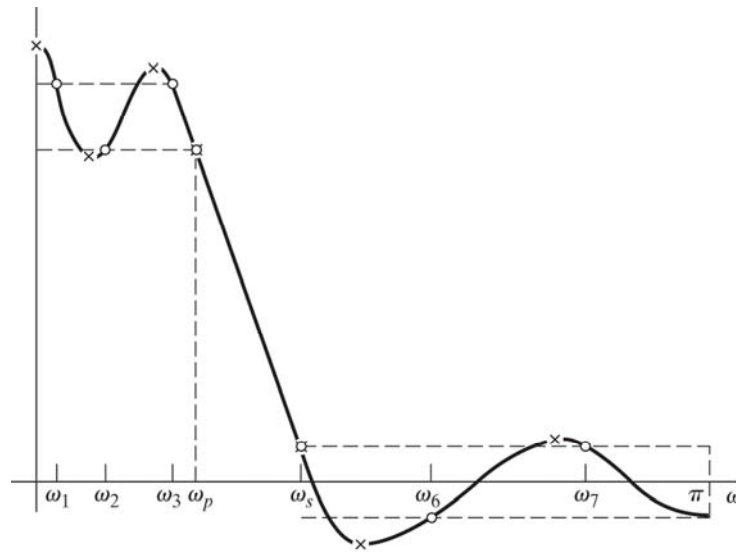
Once we know  $\{a_i\}$ , we can calculate  $A_e(e^{j\omega})$  for all  $\omega$ .

However, there is short cut. We can calculate  $A_e(e^{j\omega})$  for all  $\omega$  directly based on  $W(\omega_k)$ ,  $H_d(e^{j\omega_k})$  and  $\omega_k$  without solving for  $\{a_i\}$ .

$$A_e(e^{j\omega}) = P(\cos \omega) = \frac{\sum_{k=1}^{L+1} \left[ \frac{d_k}{(x - x_k)} \right] c_k}{\sum_{k=1}^{L+1} \left[ \frac{d_k}{(x - x_k)} \right]},$$

$$\text{where } c_k = H_d(e^{j\omega_k}) - \frac{(-1)^{k+1} \delta}{W(\omega_k)}, \quad d_k = \prod_{i=1, i \neq k}^{L+1} \frac{1}{(x_k - x_i)}$$

(Fig. 7.49, P.597)



**Figure 7.49** Illustration of the Parks-McClellan algorithm for equiripple approximation

<Flowchart (Remez algorithm)> (Fig.7.50, P.598)

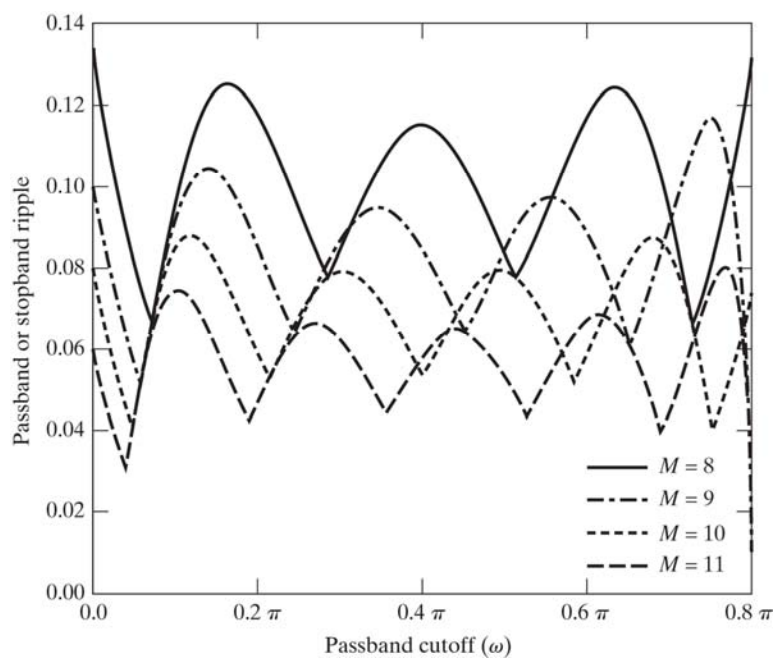
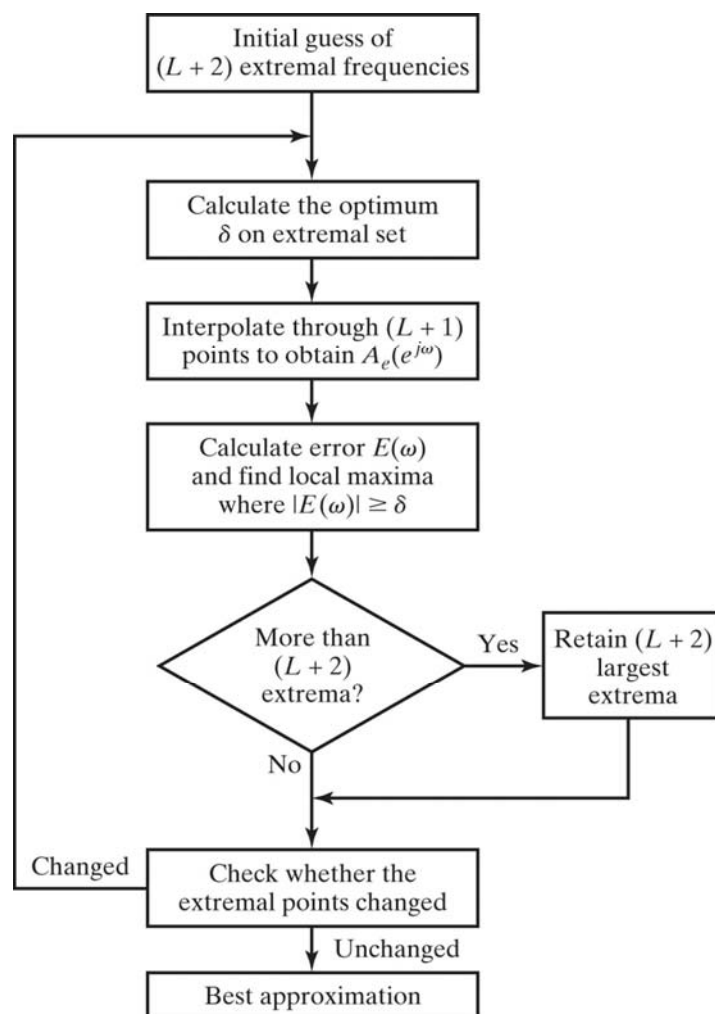


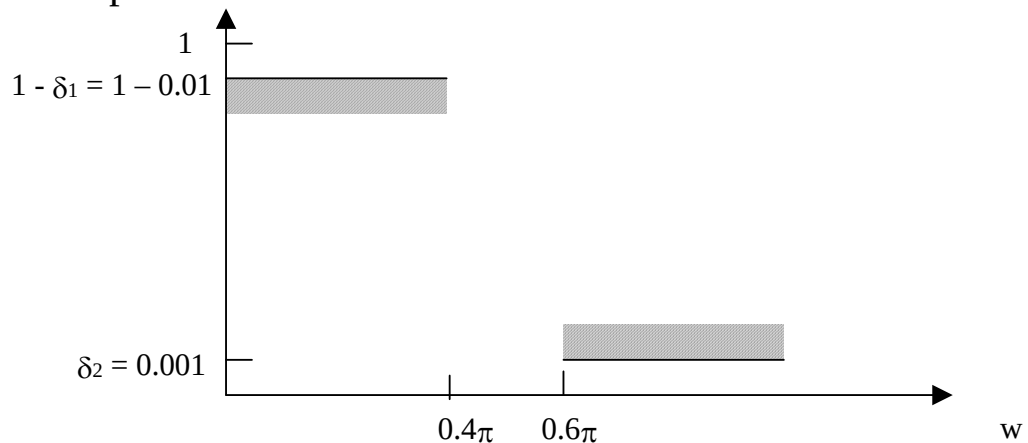
Fig. 7.51

-- How to decide  $M$  (for lowpass)? (Experimental formula)

$$M = \frac{-10\log_{10}(\delta_1\delta_2) - 13}{2.324 \cdot \Delta\omega}$$

$$\Delta\omega = \omega_s - \omega_p$$

*Example: Lowpass Filter*

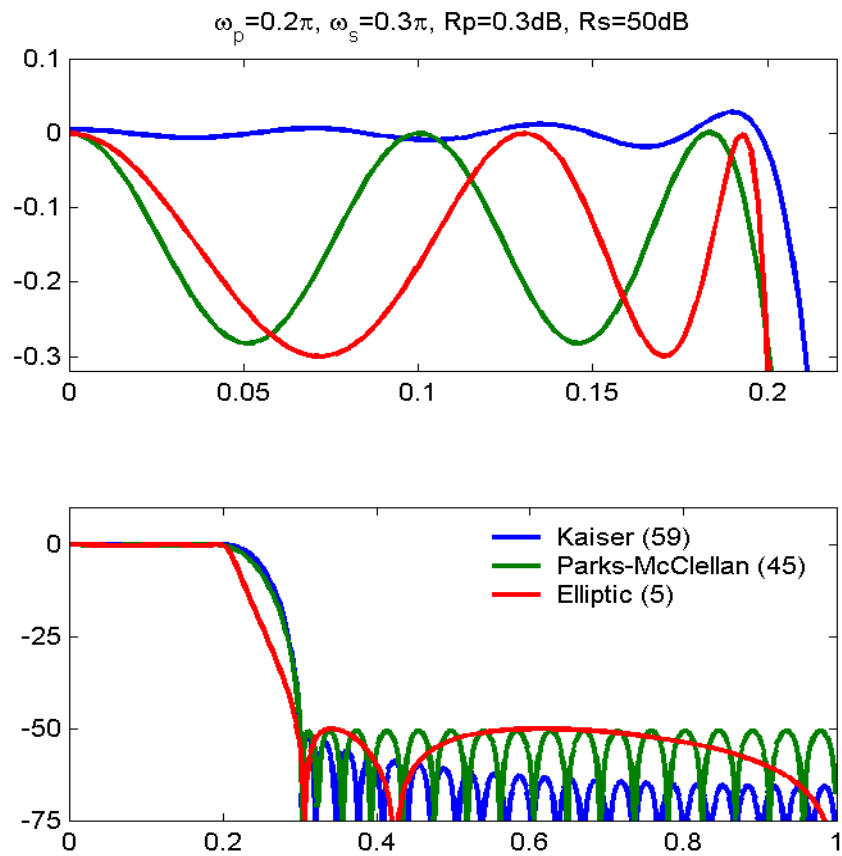
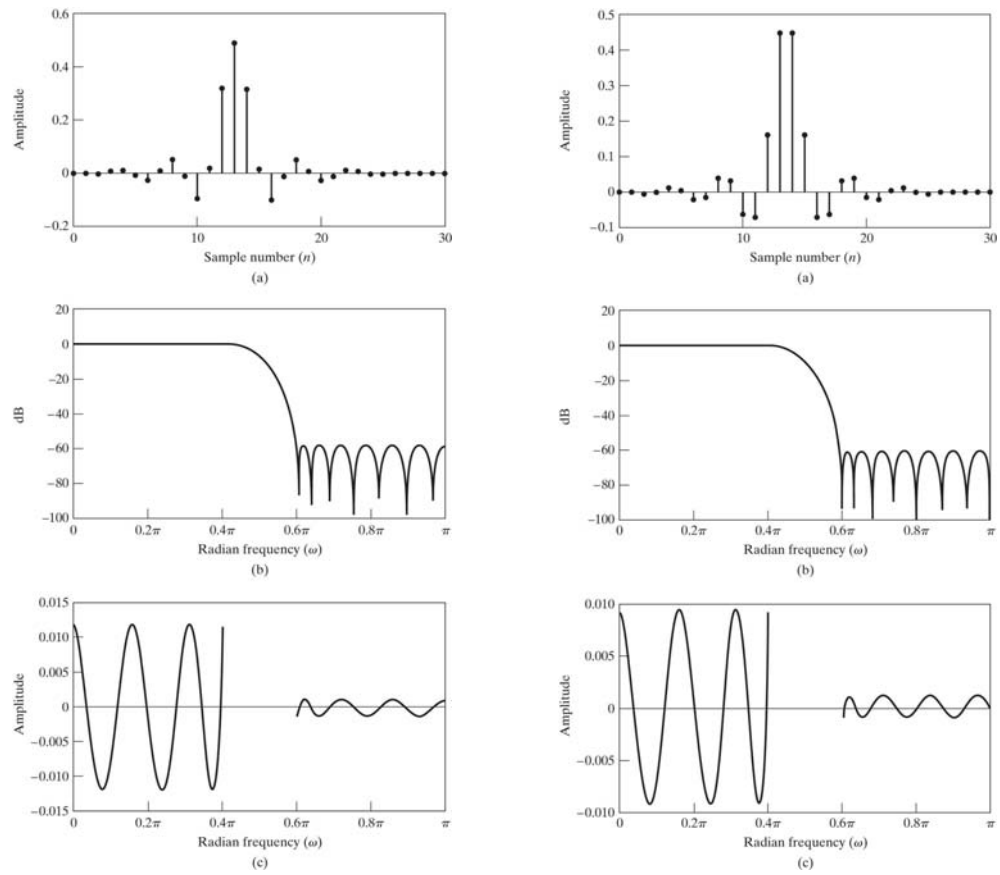


$$K = \frac{\delta_1}{\delta_2} = 10$$

$$M = \frac{-10\log_{10}(\delta_1\delta_2) - 13}{2.324 \cdot \Delta\omega} \Rightarrow M = 26$$

But the maximum errors in the passband and stopband are 0.0116 and 0.00116, respectively.  $\Rightarrow M = 27$

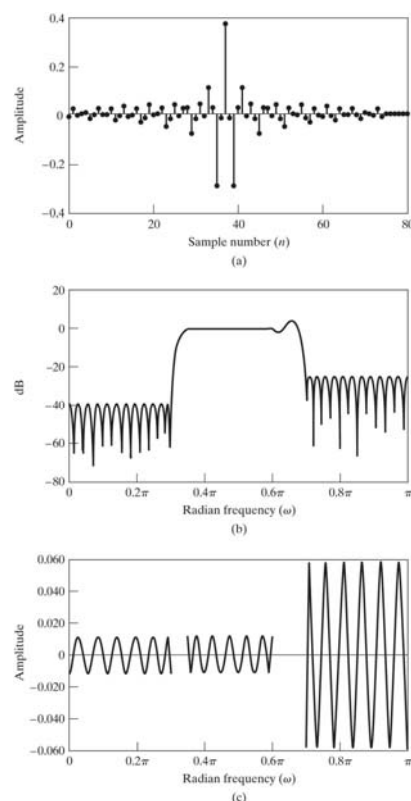
*Remark:* The Kaiser window method requires a value  $M = 38$  to meet or exceed the same specifications.



### Example: Bandpass filter

#### Remarks:

- (1) From the alternation theorem  
 $\Rightarrow$  The minimum number of alternations for the optimum approximation is  $L + 2$ .
- (2) Multiband filters can have more than  $L+3$  alternations.
- (3) Local extrema can occur in the transition regions.



| Property                                     | FIR                                | IIR                                        |
|----------------------------------------------|------------------------------------|--------------------------------------------|
| <i>Stability</i>                             | Always stable                      | Incorporate stability constraint in design |
| <i>Analog design</i>                         | No meaningful analog equivalent    | Simple transformation from analog filters  |
| <i>Phase linearity</i>                       | Can be exact linear                | Nonlinear typically                        |
| <i>Computation</i>                           | More multiplications and additions | Fewer                                      |
| <i>Storage</i>                               | More coefficients                  | Fewer                                      |
| <i>Sensitivity to coefficient inaccuracy</i> | Low sensitivity                    | Higher                                     |
| <i>Adaptivity</i>                            | Easy                               | Difficult                                  |

== End of Chapter 7 ==

- **Frequency Sampling Method** (O&S, DSP, 1975)

- In frequency domain, matches  $(M+1)$  samples of the ideal frequency response.

- Observe that  $H(e^{j\omega}) = \sum_{n=0}^M h[n] \cdot e^{-j\omega n}$

If take samples, then  $H(e^{j\omega}) \Big|_{\omega = \left(\frac{2\pi}{M+1}\right) \cdot k} = \sum_{n=0}^M h[n] \cdot e^{-j\left(\frac{2\pi}{M+1}\right) \cdot kn}$

Now, take samples of the desired model as the target,

$$\tilde{H}(k) = H_d(e^{j(2\pi/(M+1))k}), \quad k = 0, 1, \dots, M$$

The final filter impulse response is

$$h[n] = \begin{cases} \frac{1}{M+1} \sum_{k=0}^M \tilde{H}(k) \cdot e^{j\left(\frac{2\pi}{M+1}\right)k \cdot n}, & k = 0, 1, \dots, M \\ 0, & \text{otherwise} \end{cases}$$

- Take  $(M+1)$  samples of the desired  $H_d(e^{j\omega})$  and then take the inverse Fourier transform of these  $(M+1)$  samples to form  $h[n]$ .

Consequently, the FT of  $h[n]$  ( $H(e^{j\omega})$ ) would match the desired  $H_d(e^{j\omega})$  at these  $(M+1)$  sample points, BUT  $H(e^{j\omega})$  may not be

adequate at the other points  $\omega \neq \frac{2\pi}{M+1}k$ .

◀ Modify some of the  $H_d(e^{j\omega})$  sampled values to change  $H(e^{j\omega})$ . Introduce the *transition samples*.

*Remark:* Adjust the number and the values of transition samples → an optimization problem. (Rabiner et al., “An Approach to the Approximation Problem for Nonrecursive Digital Filters,” IEEE Trans. Audio Electroacoust., Vol. AU-18, June 1970, pp.83-106)